FACE DETECTION

Project work submitted to Thiruvalluvar University, Serkadu, Vellore.

In partial fulfillment for the award of degree

BACHELOR OF COMPUTER APPLICATION

BY

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DEPARTMENT OF COMPUTER APPLICATION
SHANMUGA INDUSTRIES ARTS & SCIENCE COLLEGE
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606603.

A CRM APPLICATION FOR THECHAISHOP&CO

User Story:

As a sales representative at TheChaiShop&Co, I want to efficiently manage customer relationships and streamline sales processes using our CRM application.
I want to easily access and update employer details, as well as distinguish between online and offline customer records using record types, allowing for personalized interactions.
Additionally, I need the application to automate sales calculations, such as order quantity and total bill amount, to ensure accurate and efficient transactions.
Moreover, I require the ability to generate insightful reports and dashboards to summarize and analyze sales data effectively, aiding in decision-making and strategy development.
Lastly, I would like the application to automate email notifications to customers on their birthdays, informing them of special offers tailored to celebrate their special day, enhancing customer engagement and loyalty.

1. INTRODUCTION

1.1 Project Overview :

The project “A CRM Application for TheChaiShop&Co” focuses on developing a Salesforce-based Customer Relationship Management (CRM) system to manage customer, employer, and enterprise accounts efficiently. The system automates sales processes, maintains customer records, generates reports, and improves customer engagement through automated notifications.

The project user story and objectives are defined in the initial section of the document

1.2 Purpose :

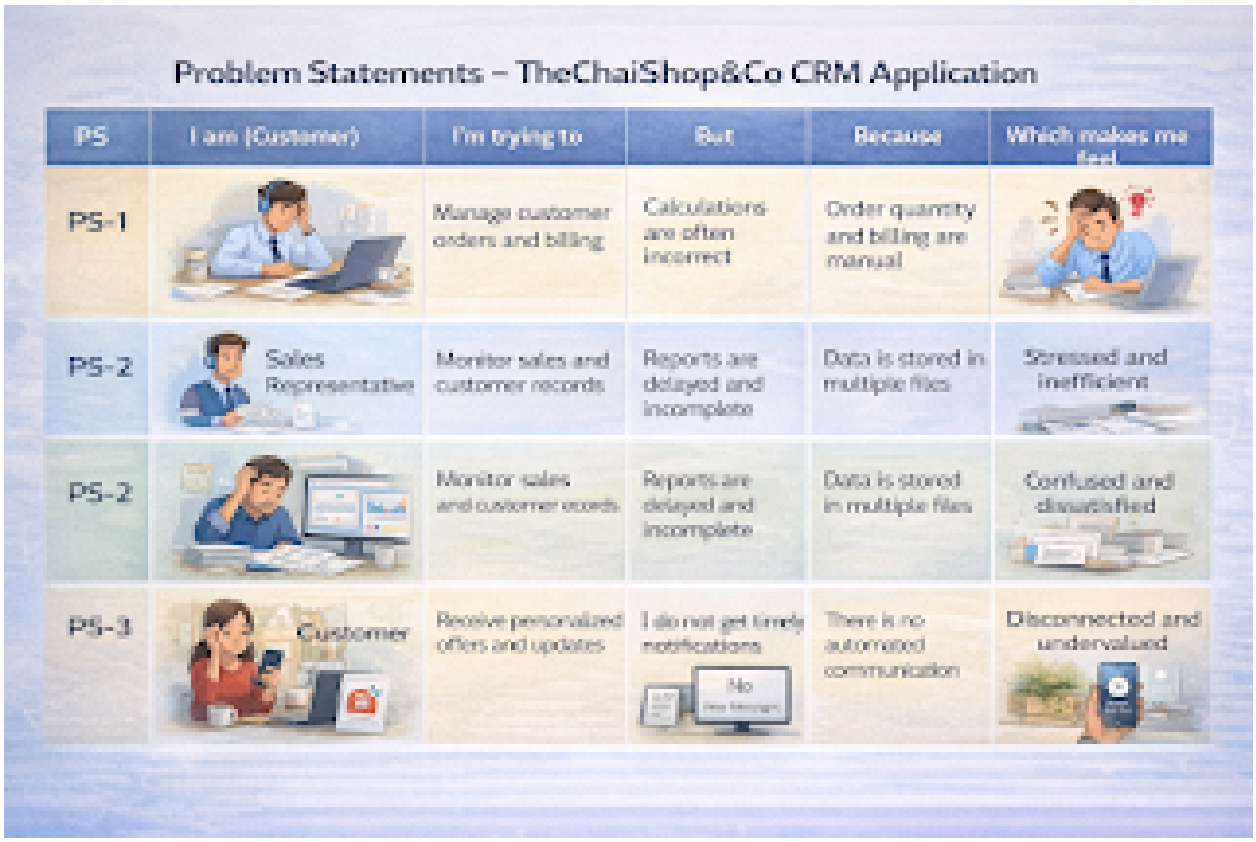
The purpose of this project is to:

- Streamline customer relationship management
- Automate billing and order calculations
- Improve data accuracy
- Enable data-driven decision-making
- Enhance customer engagement through email automation

2. IDEATION PHASE

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		billing			
PS-2	Store Manager	Monitor sales and customer records	Reports are delayed and incomplete	Data is stored in multiple files	Confused and dissatisfied
PS-3	Customer	Receive personalized offers and updates	I do not get timely notifications	There is no automated communication	Disconnected and undervalued



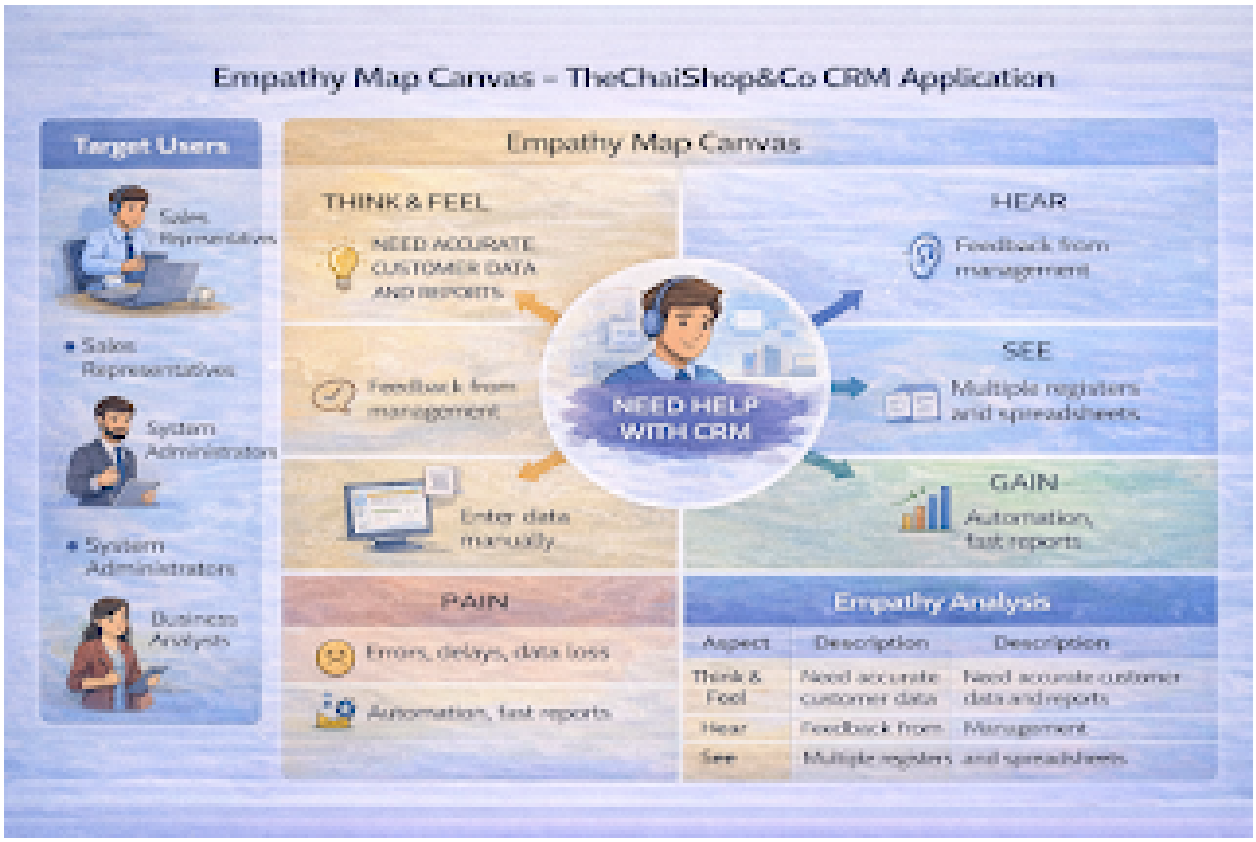
2.2 Empathy Map Canvas

Target Users

- Sales Representatives
- Managers
- System Administrators

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Think & Feel	Need accurate customer data and reports
Hear	Feedback from management
See	Multiple registers and spreadsheets
Say & Do	Enter data manually
Pain	Errors, delays, data loss
Gain	Automation, fast reports



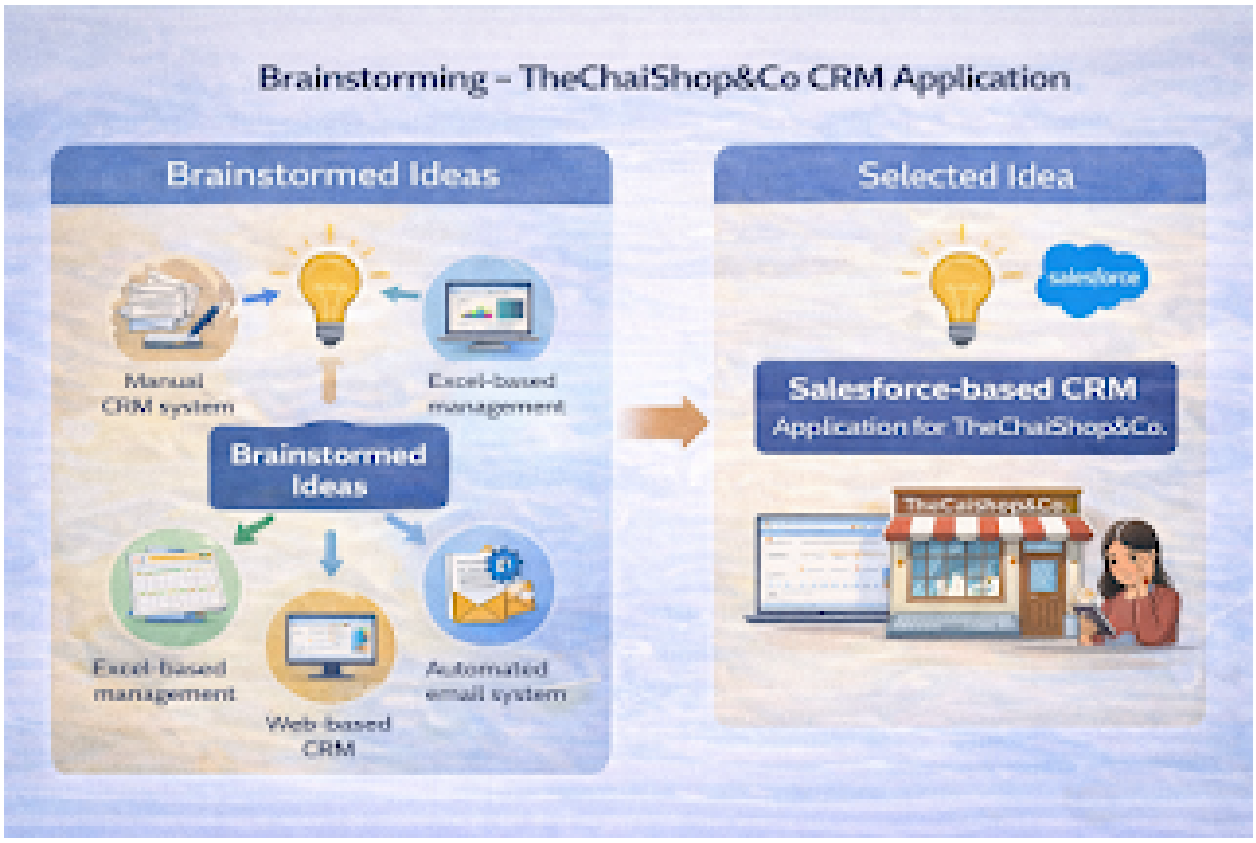
2.3 Brainstorming

Brainstormed Ideas

- Manual CRM system
- Excel-based management
- Web-based CRM

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Salesforce-based CRM Application for TheChaiShop&Co..



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

- Login
- Access CRM Application
- Enter Customer Data
- Generate Orders
- Calculate Bills

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3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Login
FR-2	Object Creation
FR-3	Data Entry
FR-4	Billing Automation
FR-5	Report Generation
FR-6	Email Automation

Non-Functional Requirements

NFR-1: Usability

The system should be easy to use for sales representatives, managers, administrators, and analysts without requiring technical training.

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- Easy-to-understand forms
- User-friendly dashboards

Example:

A new sales employee should be able to log in, enter customer details, and generate orders within a few minutes without help.

Salesforce Lightning UI provides drag-and-drop features and easy navigation.

NFR-2: Security

The system must protect customer and business data from unauthorized access.

Requirements:

- Role-based access control
- Secure login authentication
- Encrypted data storage
- Data backup and recovery

Examples:

- Profiles and permission sets
- Object-level and field-level security
- Restricted billing access
- Secure login and password policies

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- Regular system backups
- Error handling mechanisms
- Stable cloud environment

Example :

If a user enters order data and the internet disconnects, the system should save the information and restore it later.

Salesforce cloud ensures automatic backups.

NFR-4: Performance

The system should respond quickly and handle large amounts of data efficiently.

Requirements:

- Fast page loading
- Quick report generation
- Smooth dashboard rendering
- Minimal delay

Example: When a manager generates monthly sales reports for hundreds of customers, the report should load within 2–3 seconds.

Optimized queries and caching improve speed.

NFR-5: Availability

The system should be accessible whenever required.

Requirements:

- 24/7 system availability
- Minimal downtime

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Salesforce cloud provides high uptime.

NFR-6: Scalability

The system should support business growth without affecting performance.

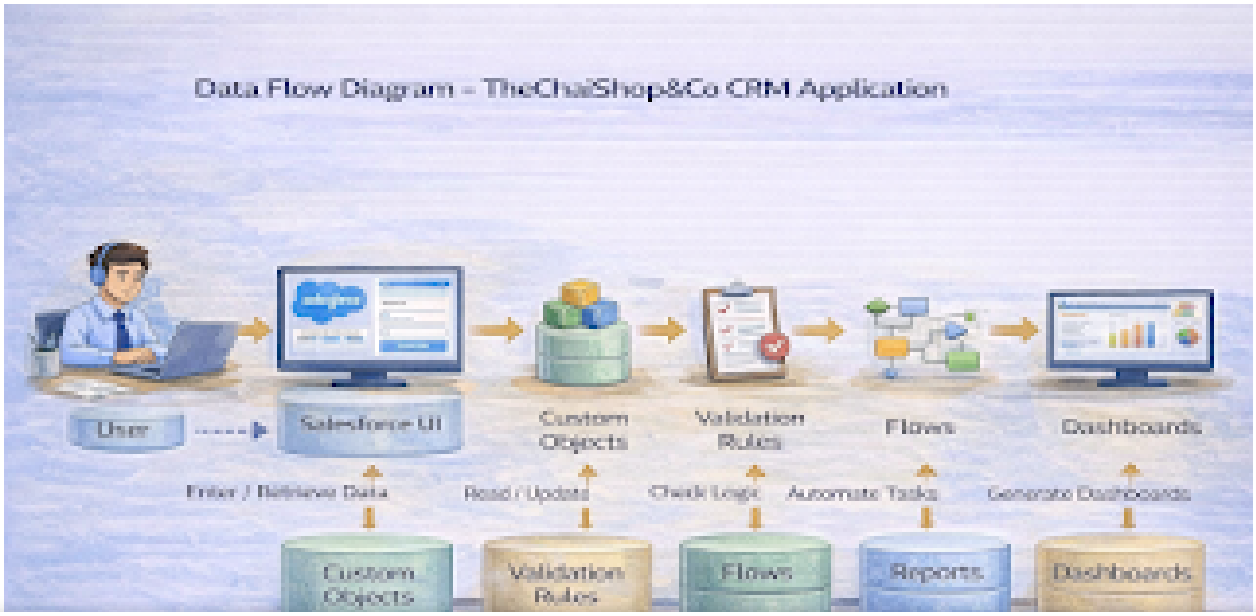
Requirements:

- Ability to handle more users
- Support increasing customer data
- Expandable storage
- Maintain system speed

Example: If TheChaiShop&Co expands to multiple branches, the CRM system should support thousands of customers without slowing down.
Salesforce infrastructure automatically scales.

3.3 Data Flow Diagram

User → Salesforce UI → Custom Objects → Validation Rules → Flows → Reports → Dashboards.



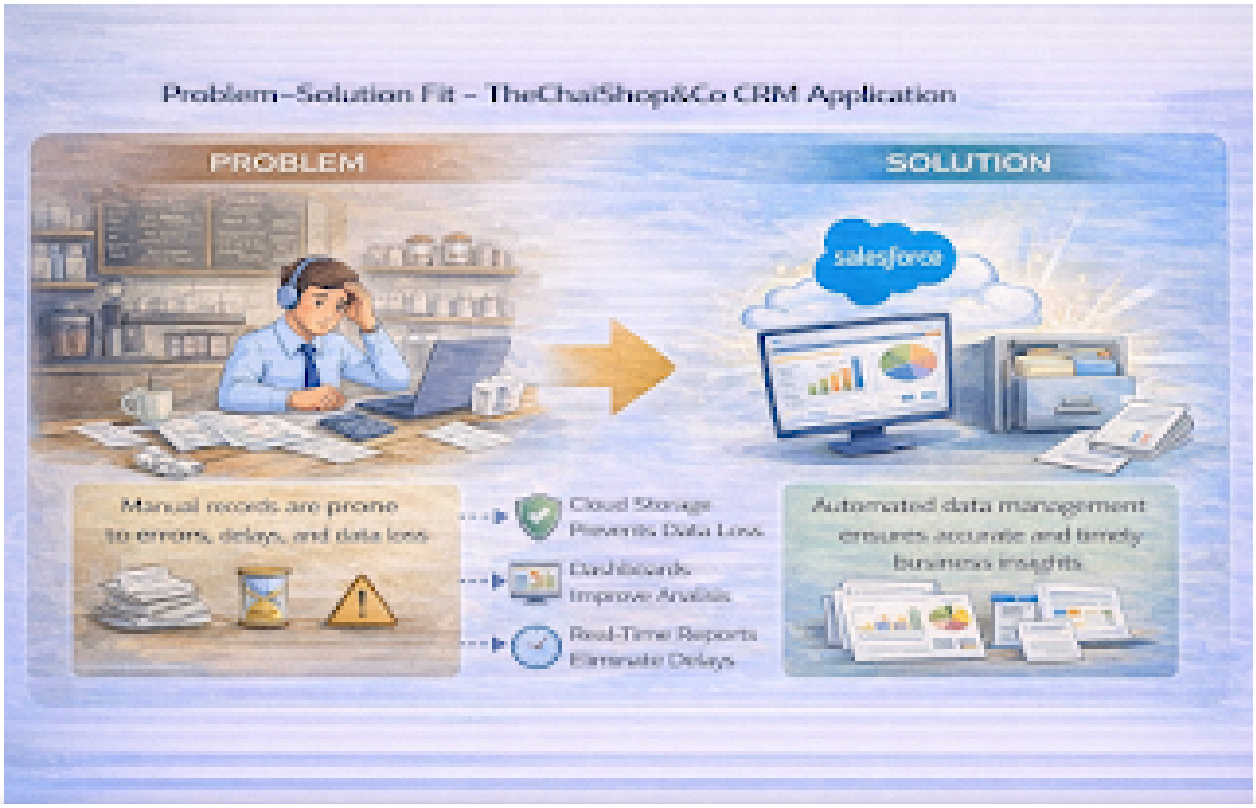
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Layer	Technology
Frontend	Salesforce Lightning
Database	Salesforce Cloud
Automation	Flows
Backend	Salesforce Platform
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The CRM system replaces manual records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely business insights generation.



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Functional Requirements

Parameter	Description
User Login & Authentication	Allows authorized users to securely log in to the CRM system using valid credentials.
Customer Record Management	Enables users to create, view, update, and delete customer details such as name, contact number, and address.
Employer & Enterprise Management	Allows management of employer and enterprise account information for business operations.
Order Management	Supports creation and tracking of customer orders, including order quantity and product details.
Automated Billing System	Automatically calculates total bill amount using formula fields based on order quantity and pricing.
Validation Rules	Ensures accurate data entry by validating fields such as phone number, PAN, and Aadhaar.
Record Types & Page Layouts	Provides different record types and layouts for various customer and enterprise categories.
Global Picklists	Maintains standardized values for fields such as product type, category, and status.
Email Notification System	Sends automated emails to customers for birthdays, offers, and order confirmations.
Workflow & Flow Automation	Automates tasks such as updating records and triggering alerts using Salesforce Flows.
Report Generation	Generates detailed reports on sales, customers, and enterprise performance.
Dashboard Visualization	Displays real-time analytics through interactive dashboards and charts.
Data Backup & Recovery	Ensures secure storage and recovery of business data in the Salesforce cloud.
User Role Management	Controls system access using profiles and permission sets for different users.

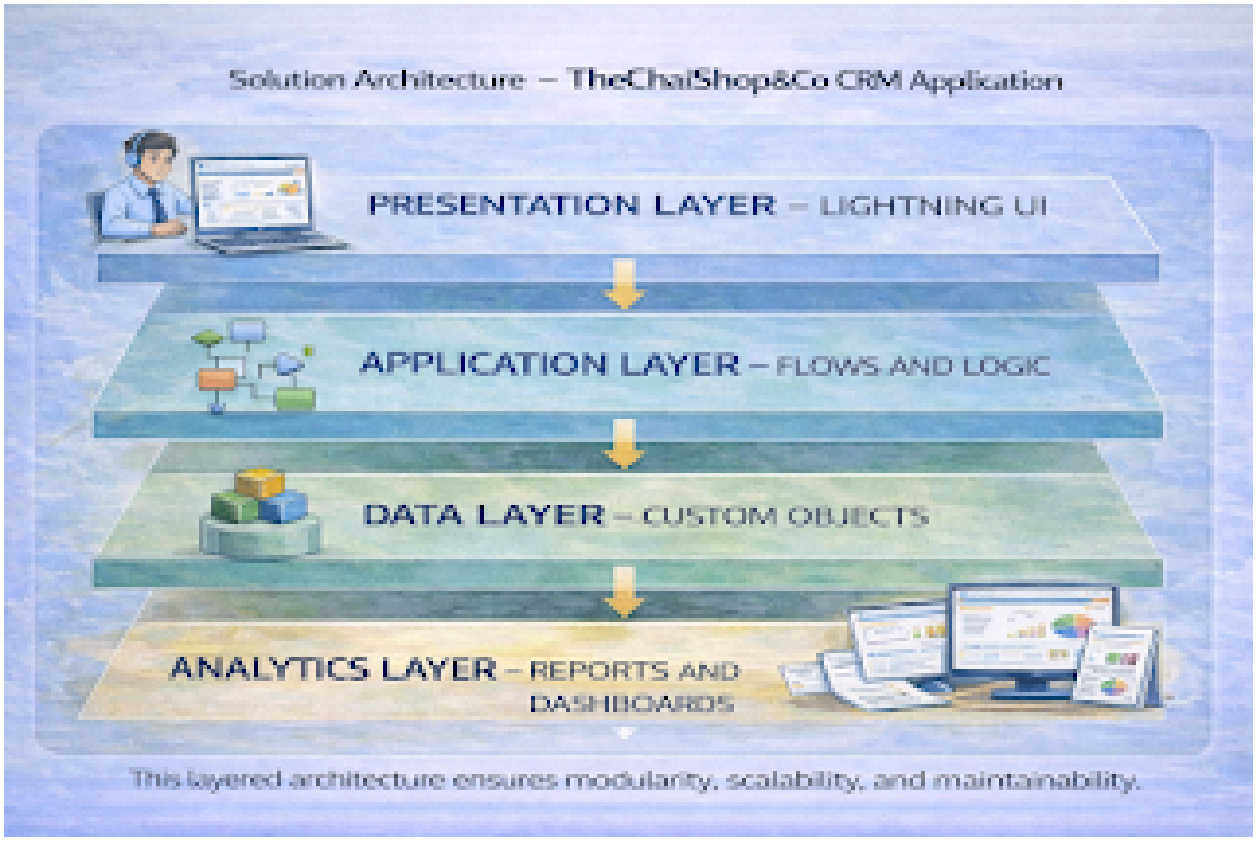
4.3 Solution Architecture

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3. Data Layer – Custom Objects

4. Analytics Layer – Reports and Dashboards

This layered architecture ensures modularity, scalability, and maintainability.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- The project followed an Agile-based development approach with defined milestones and sprints.

- A Salesforce Developer account was created to set up the development environment.

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- Page layouts and record types were created based on user roles and customer categories.
- Validation rules were implemented to prevent incorrect data entry.
- Email templates and automated flows were developed for customer notifications.
- Reports and dashboards were created for performance monitoring and analysis.
- Regular sprint reviews and testing were conducted to track progress and resolve issues.
- The project was completed within the planned timeline through systematic scheduling.

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As an admin, I want to create a Salesforce developer account so that I can configure and deploy the TheChaiShop&Co CRM application.	1	High	Member1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create Customer, Employer, and Enterprise objects so that business data can be stored and managed systematically.	3	High	Member2
Sprint-2	Data Modeling	USN-3	As a user, I want separate tabs for each object so that I can easily navigate and manage CRM records within the application.	3	High	Member3
Sprint-2	Data Modeling	USN-4	As a manager, I want a dedicated Lightning App so that all CRM operations are accessible from a single dashboard.	3	High	Member4
Sprint-2	Data Modeling	USN-5	As an admin, I want to create customer and order fields so that complete sales and billing details can be captured accurately.	5	Medium	Member1
Sprint-2	Data Modeling	USN-6	As a user, I want customized page layouts for different record types so that only relevant information is displayed during data entry	5	High	Member2
Sprint-2	Data Modeling	USN-7	As a manager, I want separate record types for Retail and Enterprise customers so that	5	Medium	Member3

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			updates and offers.			
Sprint-4	Automation	USN-10	As a system, I want to automatically calculate billing amounts so that invoices are generated without manual effort.	5	High	Member2
Sprint-5	Reports	USN-11	As a manager, I want sales and customer reports so that I can monitor business performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-12	As a manager, I want visual dashboards so that I can quickly analyze sales, customers, and enterprise performance.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Flow

stone 1 : Creation of developer account
stone 2 : Object Creation
stone 3 : Tabs
stone 4 : The Lightning App
stone 5 : Global Picklist
stone 6 : Fields and Relationships
stone 7 : Creation of Page Layout
tone 8 : Creation of Record Types

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1. Real Time Salesforce Project
2. Standard and Custom objects
3. Picklist Value sets
4. Types of Fields & their relationships.
5. Page Layouts
6. Record Types
7. Validation Rules
8. Email Templates
9. Flows
10. Reports and Dashboards.

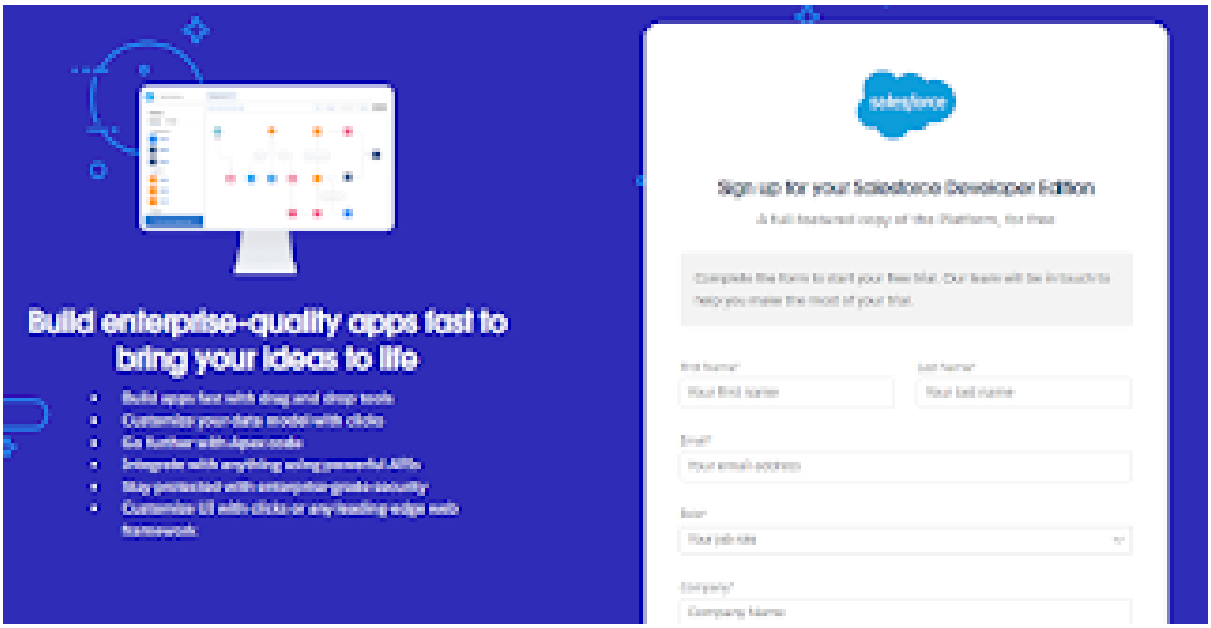
Implementation

Milestone 1 - Salesforce developer account creation :

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :

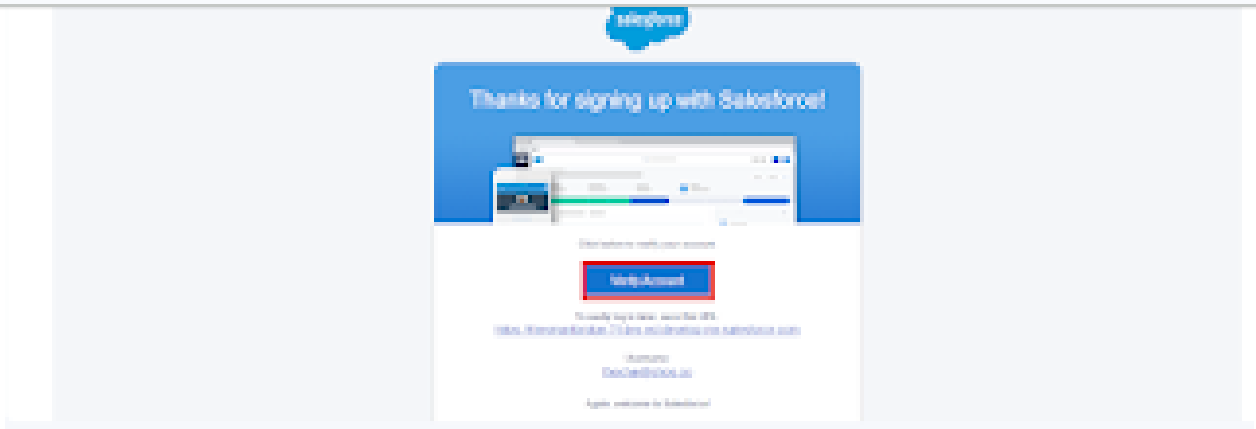


1) First name & Last name
2) Email
3) Role : Developer
4) Company : College or Company Name
5) County : India
6) Postal Code : pin code
7) Username : should be a combination of your name and company

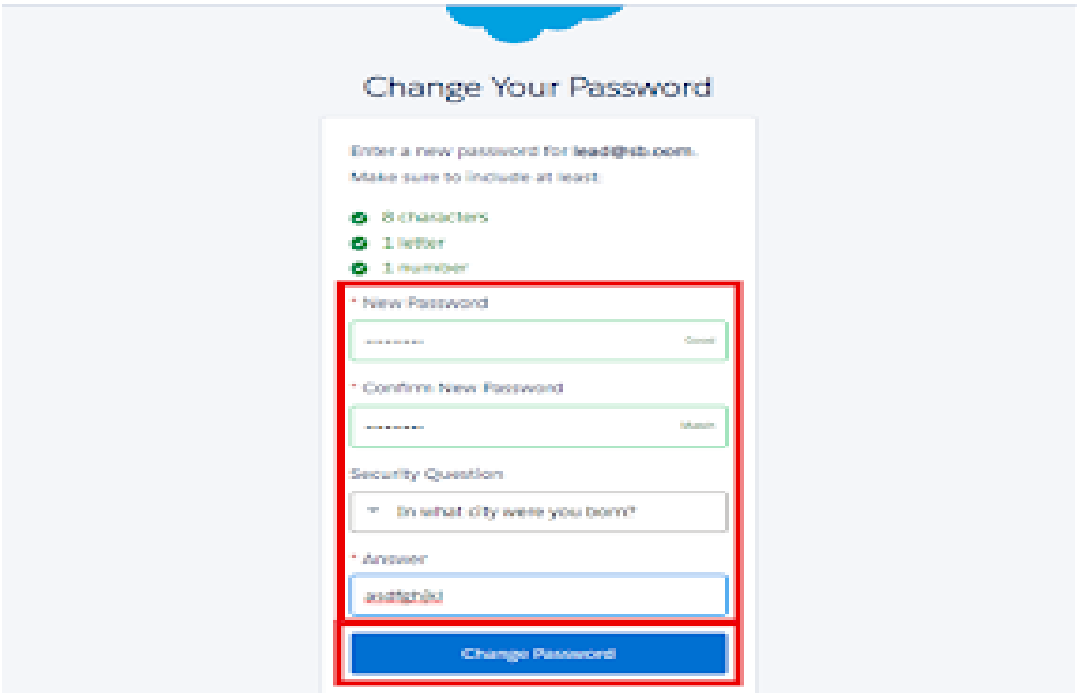
This need not be an actual email id, you can give anything in that format : username@organization.com
Click on sign me up after filling these.

Activity 2: Account Activation :

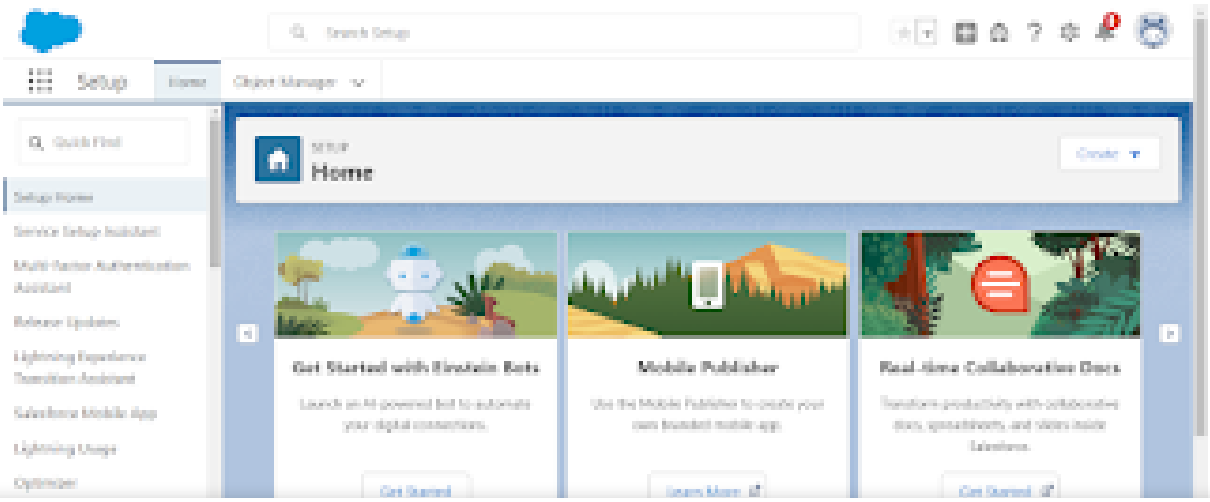
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2. Click on Verify Account.
3. Give a password and answer a security question and click on change password.

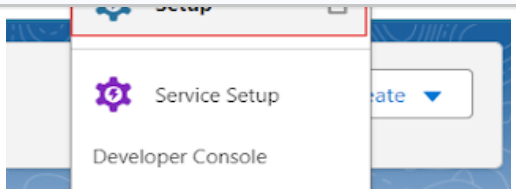


4. Then you will redirect to your salesforce setup page.

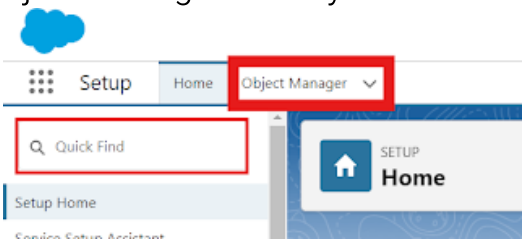


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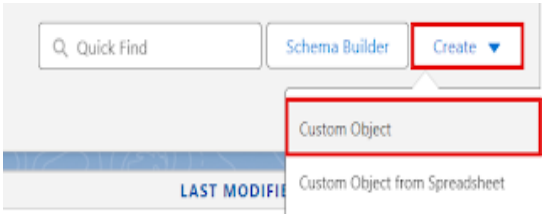
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2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.



3. On the Object Manager page, click Create | Custom Object.



Activity 1 : Employer Account Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
   - **Label** : Employer Account
   - **Plural Label** : Employer Accounts
   - Enter Record Name Label and Format
     Record Name : Employer Account Name
     Data Type **:** Text
   - **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
   - **Search Status :** check the box for Allow Search.
   - Save

Activity 2 : Customer Account Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
   - **Label** : Customer Account
   - **Plural Label** : Customer Accounts
   - Enter Record Name Label and Format
     Record Name : Customer Account Name
     Data Type **:** Text
   - **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
   - **Search Status :** check the box for Allow Search.
   - Save

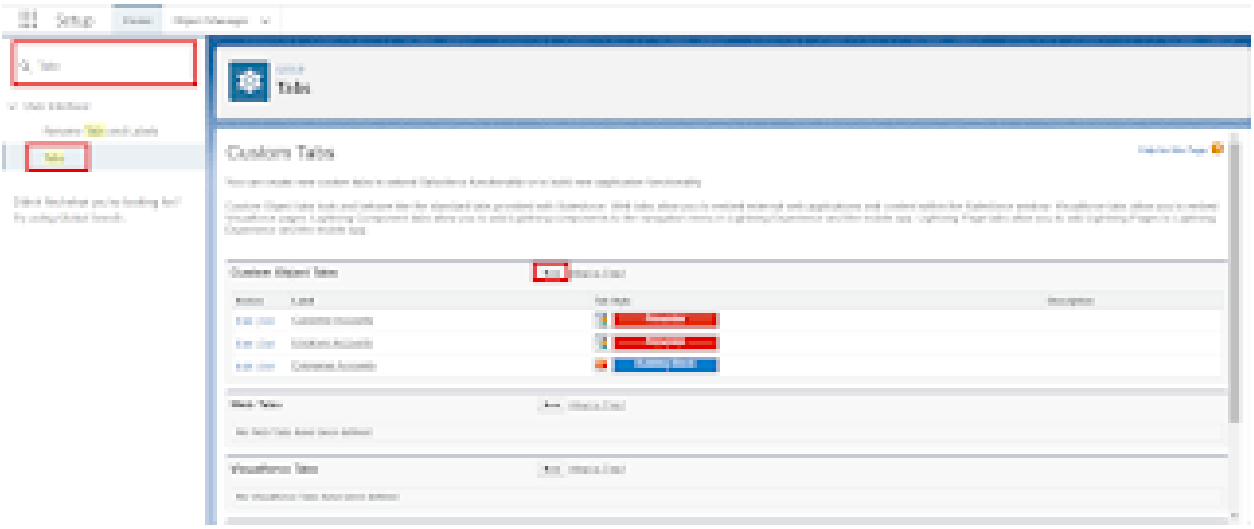
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- Save

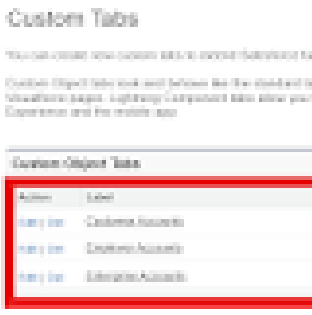
Milestone 3 - Tabs

Activity 1 : Custom tabs creation

1. Go to setup page → type Tabs in Quick Find bar → click on tabs → New (under custom object tab)



2. Select Object(Employer Accounts) → Select the tab style → Next (Add to profiles page) keep it as default → Next (Add to Custom App) uncheck the include tab .
3. Make sure that the Append tab to users' existing personal customizations is checked.
4. Click save.
5. Repeat the same steps for Customer Accounts and Enterprise Accounts.

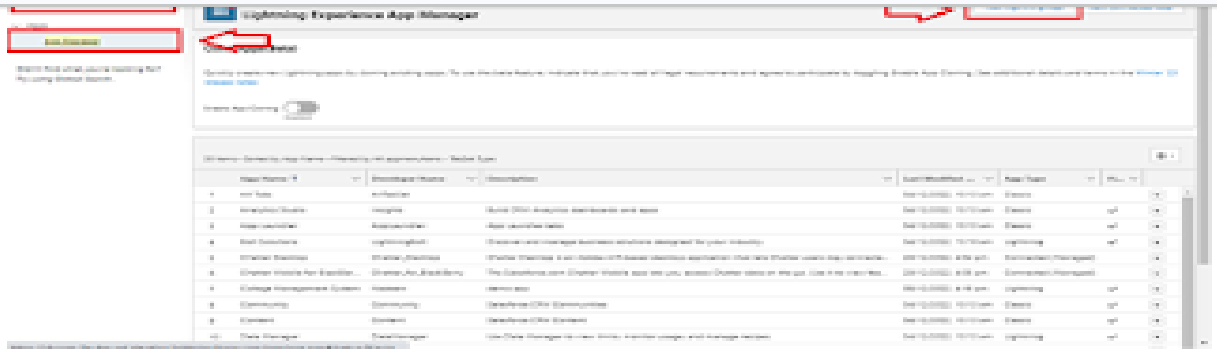


Milestone 4 - The Lightning App

Activity 1 : Create a lightning app

1. Go to setup page → search “app manager” in quick find → select “app manager” → click on New lightning App.

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2. Fill the app name in app details and branding as follow
App Name : TheChaiShop&Co
Developer Name : Auto Populates
Add image optional (if you want to give any image you can otherwise not mandatory) - Add Primary color Hex or leave it to default.
Then click Next.

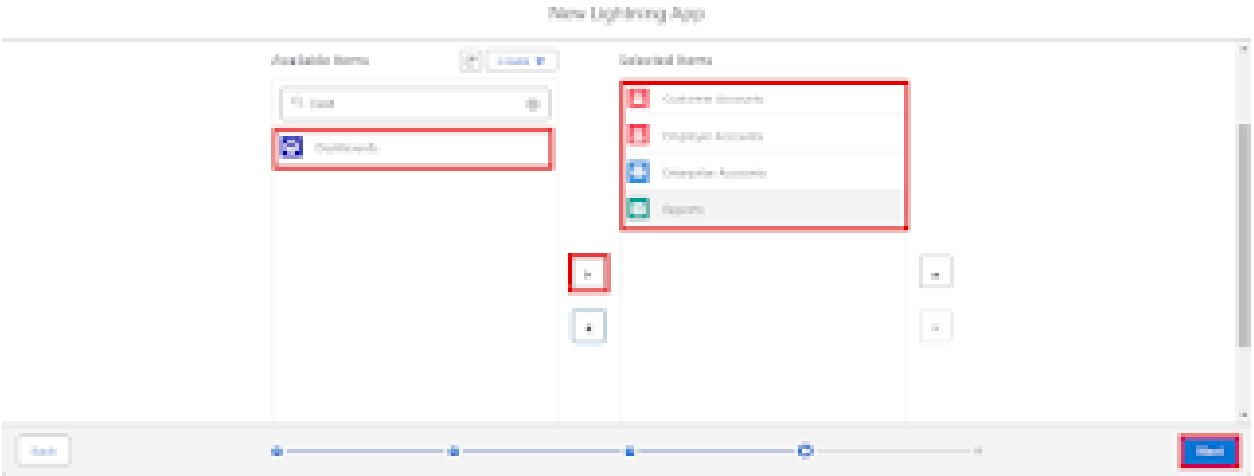
3. In App options
Navigation style : Standard navigation
Setup experience : setup
Supporters form factors : Desktop and phone
Then click Next

4. In Utility items
Utility Bar alignment : Default
Then click Next.

5. Navigation items
Select the created Custom Objects and required standard objects

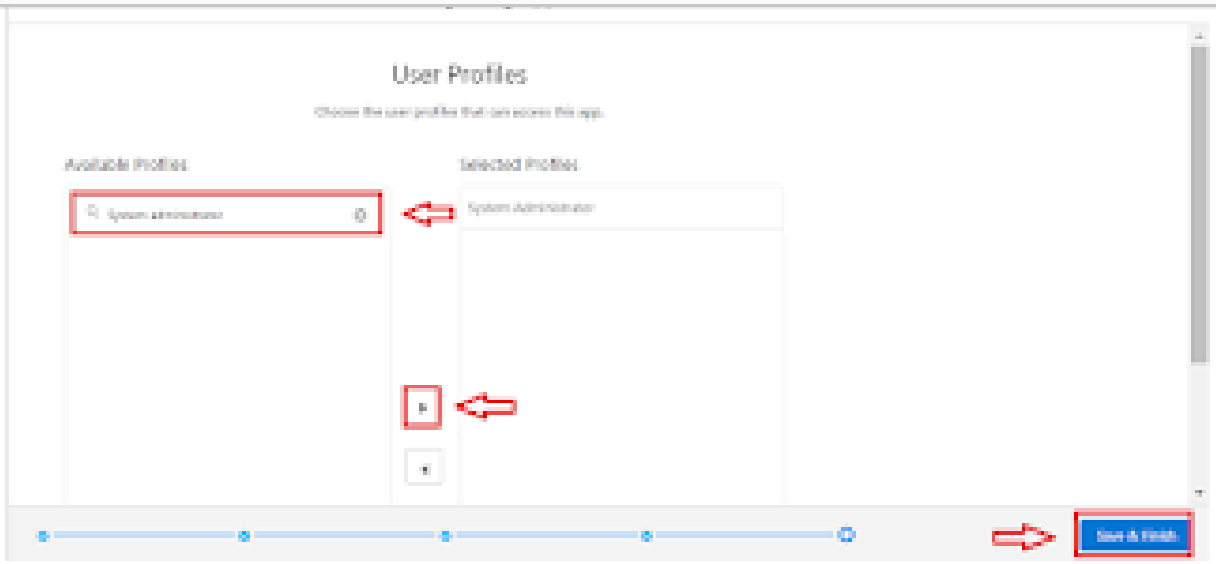
- Customer Accounts

Employer Accounts
Enterprise Accounts
Reports
Dashboards



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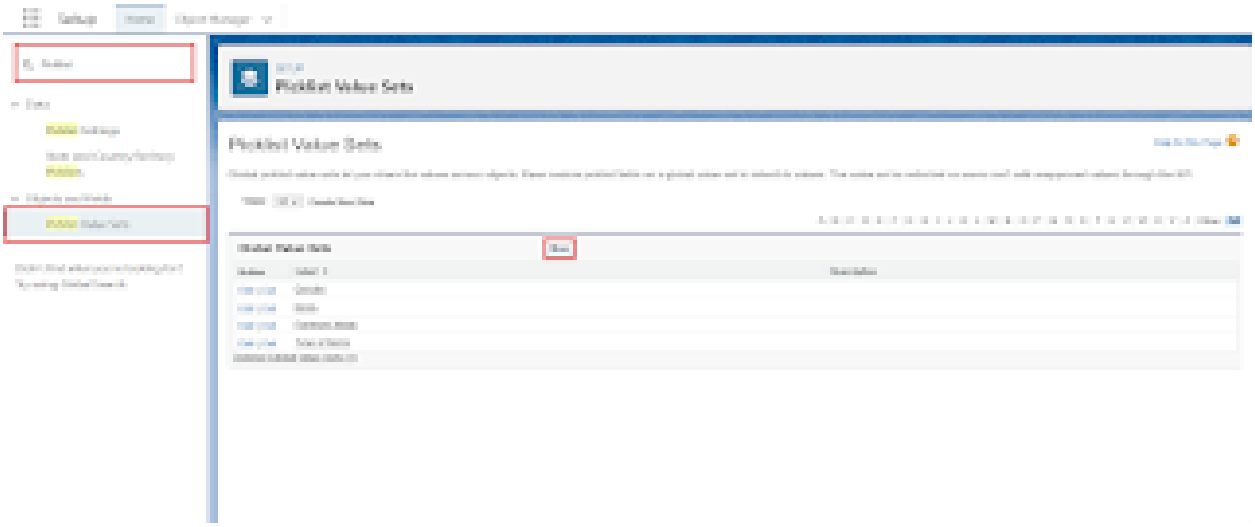


7. Click Save & Finish.

Milestone 5 : Global Picklist

Activity 1 : Create Global PIcklist

1. In Quick find box search for : Picklist value sets and click New.



• **Label** : Gender
Name : Auto Populates
Values : Male
Female
Others
Click **Save & New.**



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Values : Platinum Tea Powder
Double Delight Tea Powder
Super Dust Tea Powder
Osmania Biscuits (Value Pack)
Osmania Biscuits (Premium Pack)
Dry Fruit Biscuits
Fruit Biscuits
Salt Biscuits
Multigrain Biscuits
Ragi Biscuits
Double Choco Chip Cookies
Butter Almond Cookies
Melting Moments Cookies
Honey Oats Cookies
Delight Box
Signature Box
Assorted Box

Click **Save & New.**

- **Label** : Payment Mode

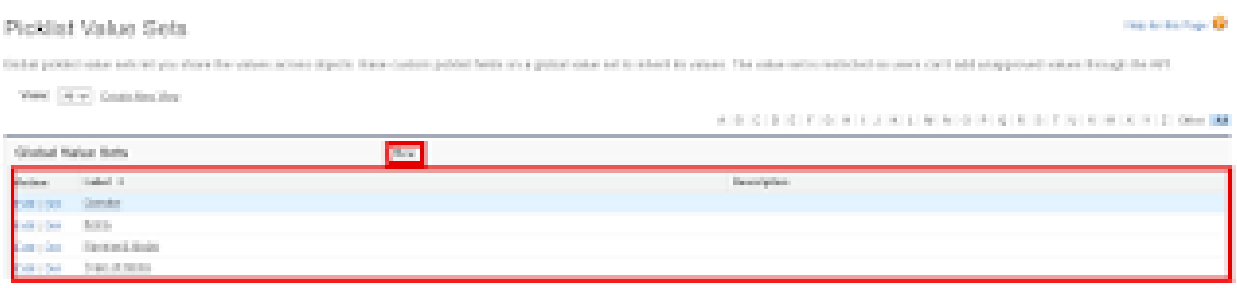
Name : Auto populates
Values : Gpay
PhonePe
Paytm
Credit or Debit card
Cash.

Click **Save & New.**

- **Label :** Type of Items

Name : Auto populates
Values : Tea powder
Cookies
Gift Hampers

Click **Save.**

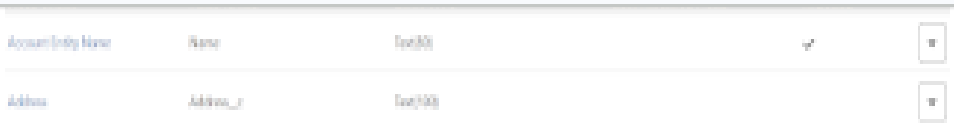


Milestone 6 : Fields & Relationships

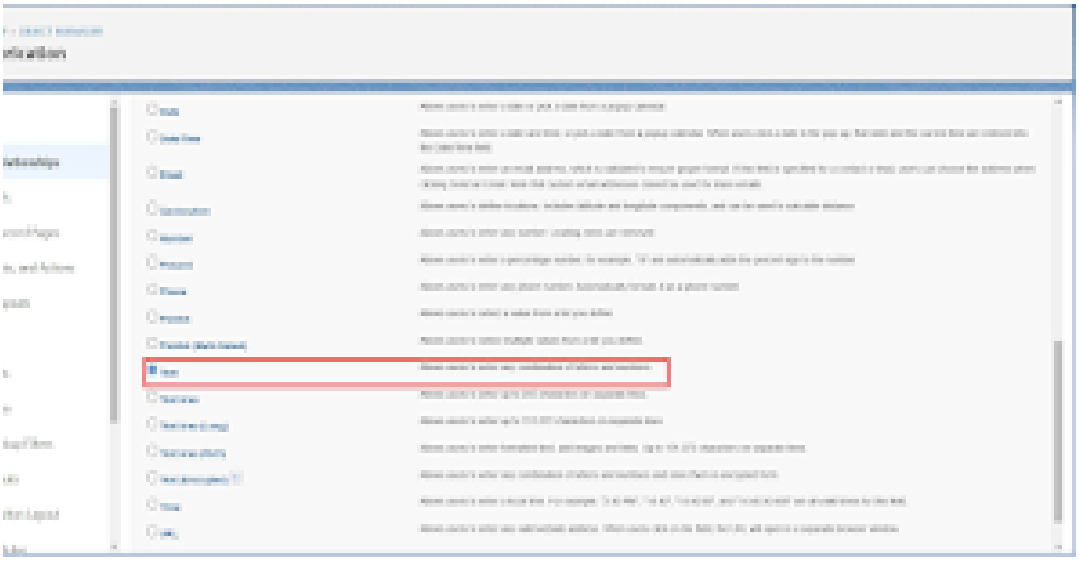
Activity 1 : Create Custom fields for Employer Account

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup

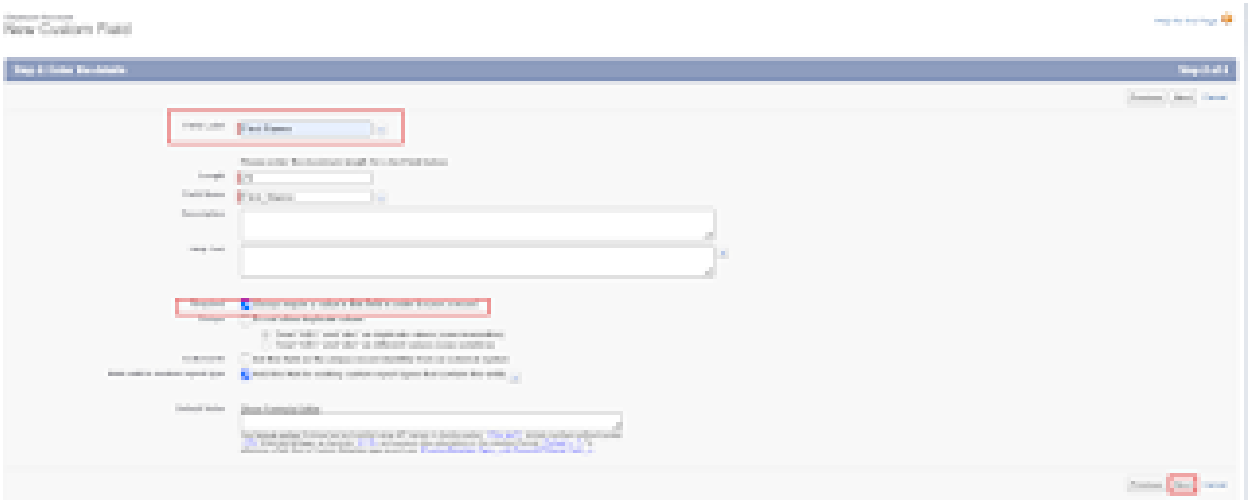
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6. Next, choose a data type as Text. Choosing a data type helps you format the field input.



7. Click on next.



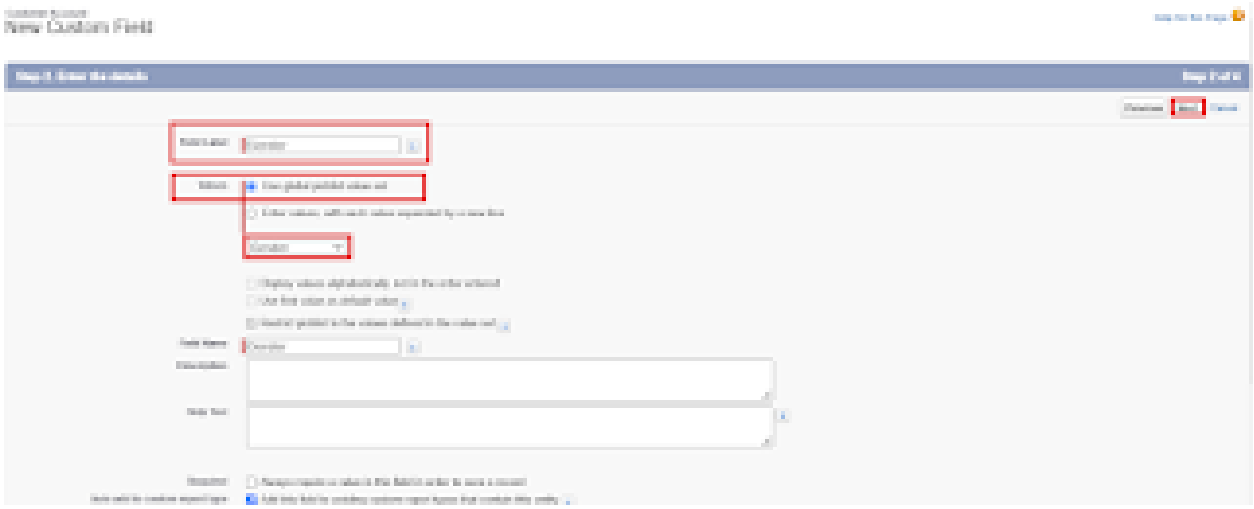
8. Data Type : Text
Field Label: First Name
Field Name : Auto Populates
Length : 25
Required : Check the box
Click on Next → Next → Save and new.

9. Repeat the Same steps for remaining fields

- Data Type : Text
Field Label: Last Name
Field Name : Auto Populates

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• Data Type : Phone
Field Label : Phone Number
Field Name : Auto Populates
Required : Check the box
Click on Next → Next → Save and new.

• Data Type : Email
Field Label : Email ID
Field Name : Auto Populates
Required : Check the box
Click on Next → Next → Save and new.

• Data Type : Date
Field Label : Date of Birth
Field Name : Auto Populates
Required : check the box
Click on Next → Next → Save and new.

• Data Type : Text
Field Label : Aadhar Number
Field Name : Auto Populates
Length : 12
Click on Next → Next → Save and new.

• Data Type : Text
Field Label : PAN Details
Field Name : PAN_Details
Length : 10
Click on Next → Next → Save and new.

• Data Type : **Picklist**
Field Label : Designation
Field Name : Auto Populates
Value : Enter value, with each value separated by a new line.

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Activity 2: Create Custom fields for Customer Account

1. Data Type : Text
Field Label: First Name
Field Name : Auto Populates
Length : 25
Required : Check the box
Click on Next → Next → Save and new.

2. Data Type : Text
Field Label: Last Name
Field Name : Auto Populates
Length : 25
Required : Check the box
Click on Next → Next → Save and new.

3. Data Type : **PickList**
Field Label: Gender
Values : Use **Global Picklist** value set and select Gender from the dropdown menu.
Field Name : Auto Populates.
Required : Check the box
Click on Next → Next → Save and new.

4. Data Type : Phone
Field Label : Phone Number
Field Name : Auto Populates
Required : Check the box
Click on Next → Next → Save and new.

5. Data Type : Date
Field Label : Date of Birth
Field Name : Auto Populates
Required : check the box
Click on Next → Next → Save and new.

6. Data Type : **Formula**
Field Label : Date of Birthday is today ?
Field Name : Auto Populates
Formula Return : **Checkbox**
Formula :

```
IF(
AND(
DAY( TODAY() ) = DAY( Date_of_Birth__c ) ,
MONTH ( TODAY()) = MONTH( Date_of_Birth__c )),
TRUE,
FALSE)
```

Click on Next → Next → Save and new.

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Click on Next → Next → Save and new.

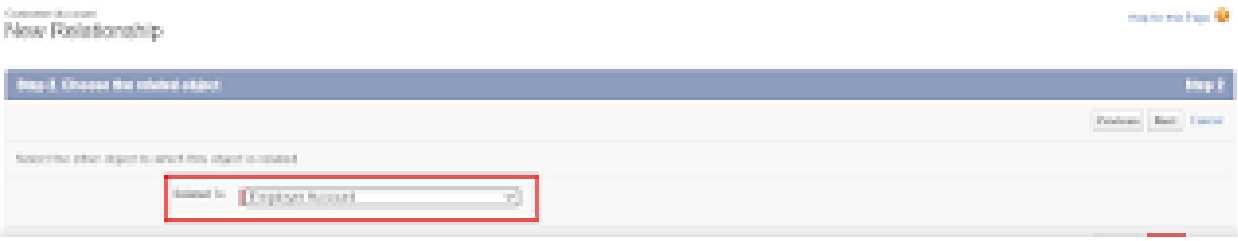
9. Date Type : **Multi Select Picklist**
Field Label : Type of Items
Values : Use **Global Picklist** value set and
select **Type of Items** from the dropdown menu.
Field Name : Auto Populates.
Required : Check the box
Click on Next → Next → Save and new.

10. Date Type : **Multi Select Picklist**
Field Label : Products purchased
Values : Use **Global Picklist** value set and
select **Items** from the dropdown menu.
Field Name : Auto Populates.
Required : Check the box
Click on Next → Next → Save and new.

11. Date Type : **Multi Select Picklist**
Field Label : Chai
Value : Enter value, with each value separated by a new line.
TheChaiShop&Co Special Tea
Single Tea
Masala Tea
Lemon Tea
Ginger Tea
Green Tea
Sugar Free Tea
Click on Next → Next → Save and new.

12. Data Type : **Picklist**
Field Label : Payment Mode
Values : Use **Global Picklist** value set and
select **Payment Mode** from the dropdown menu.
Field Name : Auto Populates.
Required : Check the box
Click on Next → Next → Save and new.

13. Data Type : **Lookup Relationship**
Related To : Employer Account
Field Label : Associated Employer
Field Name : Auto Populates
Click on Next → Next → Save and new.

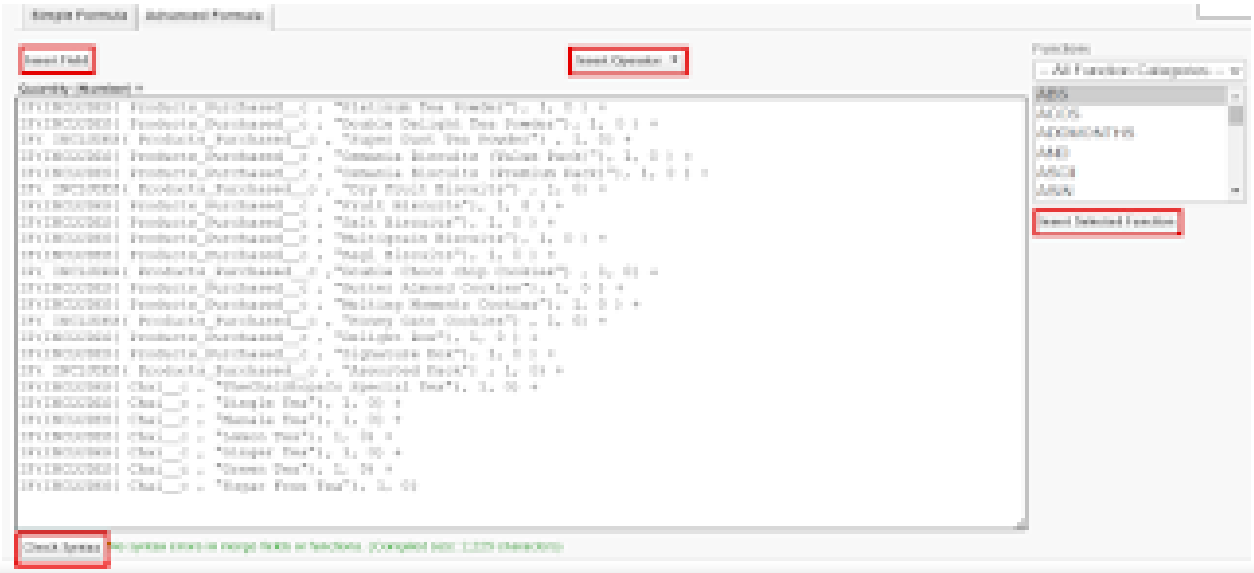


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Enter Formula = Quantity (Number) =

```
IF(INCLUDES( Products_Purchased__c , "Platinum Tea Powder"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Double Delight Tea Powder"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Super Dust Tea Powder") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Value Pack)"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Dry Fruit Biscuits") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Fruit Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Salt Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Multigrain Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Ragi Biscuits"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c ,"Double Choco chip Cookies") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Butter Almond Cookies"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Melting Moments Cookies"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Honey Oats Cookies") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Delight Box"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Signature Box"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Assorted Pack") , 1, 0) +
        IF(INCLUDES( Chai__c , "TheChaiShop&Co Special Tea"), 1, 0) +
        IF(INCLUDES( Chai__c , "Single Tea"), 1, 0) +
        IF(INCLUDES( Chai__c , "Masala Tea"), 1, 0) +
        IF(INCLUDES( Chai__c , "Lemon Tea"), 1, 0) +
        IF(INCLUDES( Chai__c , "Ginger Tea"), 1, 0) +
        IF(INCLUDES( Chai__c , "Green Tea"), 1, 0) +
        IF(INCLUDES( Chai__c , "Sugar Free Tea"), 1, 0)
```

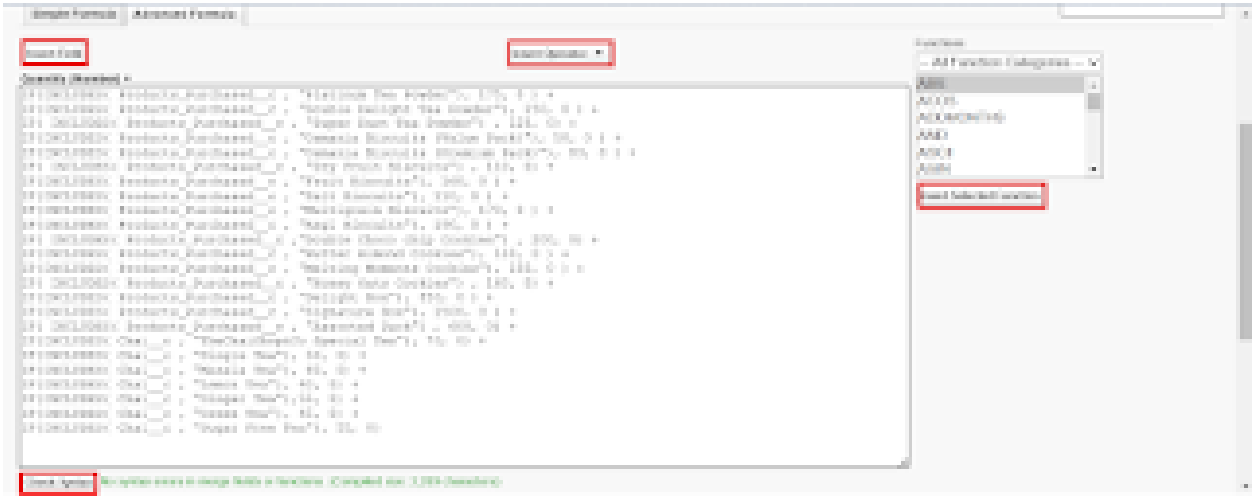


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```
IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 80, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Dry Fruit Biscuits") , 110, 0) +
IF(INCLUDES( Products_Purchased__c , "Fruit Biscuits"), 160, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Salt Biscuits"), 110, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Multigrain Biscuits"), 170, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Ragi Biscuits"), 190, 0 ) +
IF( INCLUDES( Products_Purchased__c ,"Double Choco chip Cookies") , 200, 0) +
IF(INCLUDES( Products_Purchased__c , "Butter Almond Cookies"), 110, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Melting Moments Cookies"), 150, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Honey Oats Cookies") , 160, 0) +
IF(INCLUDES( Products_Purchased__c , "Delight Box"), 850, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Signature Box"), 1500, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Assorted Pack") , 600, 0) +
IF(INCLUDES( Chai__c , "TheChaiShop&Co Special Tea"), 70, 0) +
IF(INCLUDES( Chai__c , "Single Tea"), 50, 0) +
IF(INCLUDES( Chai__c , "Masala Tea"), 60, 0) +
IF(INCLUDES( Chai__c , "Lemon Tea"), 40, 0) +
IF(INCLUDES( Chai__c , "Ginger Tea"),55, 0) +
IF(INCLUDES( Chai__c , "Green Tea"), 45, 0) +
IF(INCLUDES( Chai__c , "Sugar Free Tea"), 55, 0)
```



Click on Next → Next → Save.

Activity 3: Create Custom fields for Enterprise Account

1. Data Type : Text
Field Label: Enterprise Name
Field Name : Auto Populates
Length : 25
Required : Check the box
Click on Next → Next → Save and new.

2. Data Type : Phone
Field Label : Phone Number
Field Name : Auto Populates
Required : Check the box
Click on Next → Next → Save and new.

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Click on Next → Next → Save and new.

5. Date Type : **Multi Select Picklist**
Field Label : Type of Items
Values : Use **Global Picklist** value set and
select **Type of Items** from the dropdown menu.
Field Name : Auto Populates.
Required : Check the box
Click on Next → Next → Save and new.

6. Date Type : **Multi Select Picklist**
Field Label : Products purchased
Values : Use **Global Picklist** value set and
select **Items** from the dropdown menu.
Field Name : Auto Populates.
Required : Check the box
Click on Next → Next → Save and new.

7. Data Type : **Lookup Relationship**
Related To : Employer Account
Field Label : Associated Employer
Field Name : Auto Populates
Click on Next → Next → Save and new.

8. Data Type : **Formula**
Field Label : Order Quantity
Field Name : Auto Populates
Formula Return : **Number**
Enter Formula = Quantity (Number) =

```
IF(INCLUDES( Products_Purchased__c , "Platinum Tea Powder"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Double Delight Tea Powder"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Super Dust Tea Powder") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Value Pack)"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Dry Fruit Biscuits") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Fruit Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Salt Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Multigrain Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Ragi Biscuits"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c ,"Double Choco chip Cookies") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Butter Almond Cookies"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Melting Moments Cookies"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Honey Oats Cookies") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Delight Box"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Signature Box"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Assorted Pack") , 1, 0)
```

Click on Next → Next → Save and new.

9. Data Type : **Formula**
Field Label : Total Bill
Field Name : Auto Populates
Formula Return : **Number**

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```
IF( INCLUDES( Products_Purchased__c ,'Double Choco chip Cookies') , 200, 0) +
IF(INCLUDES( Products_Purchased__c , "Butter Almond Cookies"), 110, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Melting Moments Cookies"), 150, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Honey Oats Cookies") , 160, 0) +
IF(INCLUDES( Products_Purchased__c , "Delight Box"), 850, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Signature Box"), 1500, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Assorted Pack") , 600, 0)
```

10. Data Type : **Picklist**
Field Label : Payment Mode
Values : Use **Global Picklist** value set and
select **Payment Mode** from the dropdown menu.
Field Name : Auto Populates.
Required : Check the box
Click on Next → Next → Save.

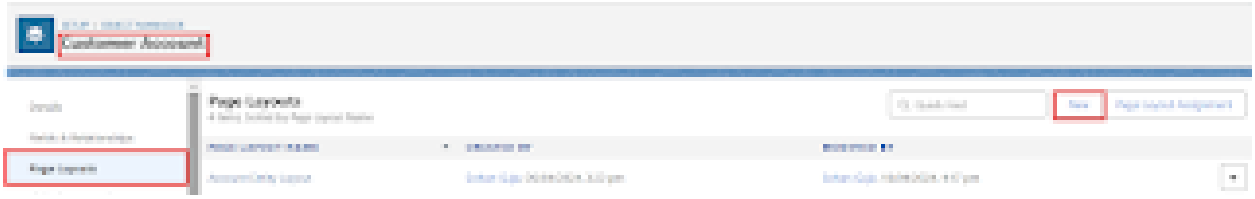
Milestone 7 : Page Layouts

For the personalized field view in Customer Accounts, let's create two page layouts Online orders and Walk-In orders.

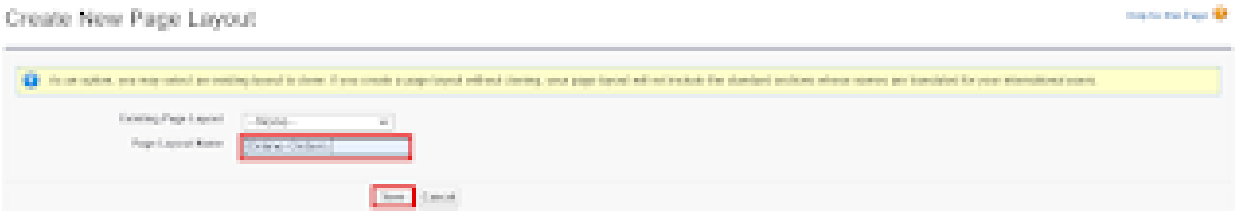
Activity 1: Online Orders page layout

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

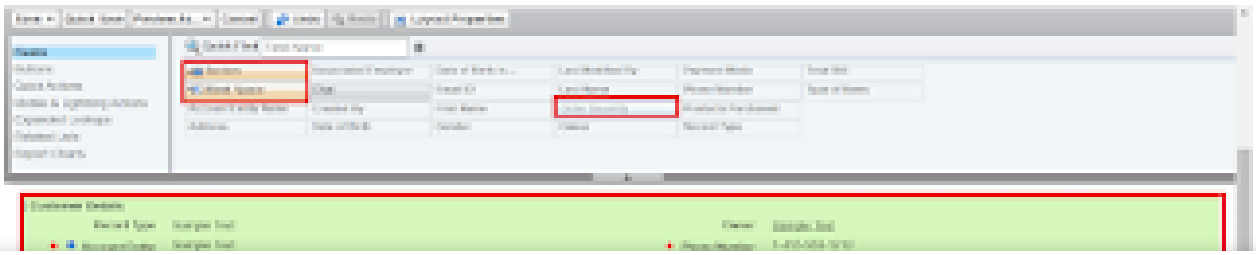
2. Click on the Page Layouts → click New.



3. Page Layout Name : Online Orders and click Save.



4. Drag and Arrange all the required fields as shown below.



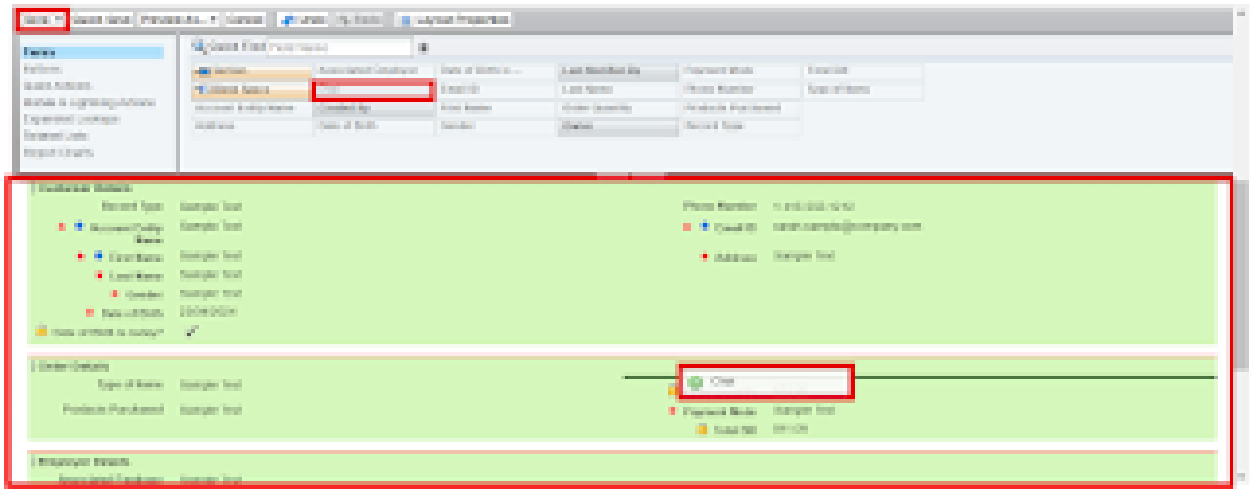
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1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the Page Layouts → click New.

3. Page Layout Name : Walk-In orders and click save.

4. Drag and Arrange all the required fields as shown below.



5. Click **Save**.

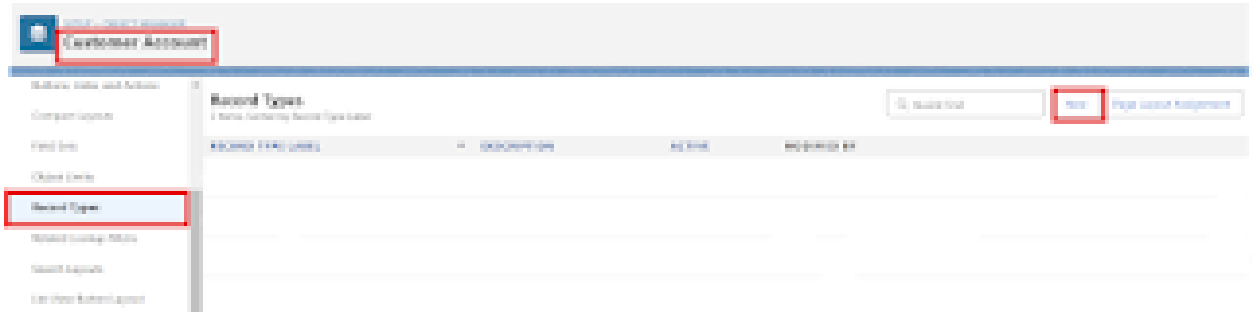
Milestone 8 - Record Types

To Differentiate the records and data stored in Customer Accounts, let's create two different Record Types. Online orders and Walk-In orders.

Activity 1: Online Orders Record Type

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the Record Types → click New.



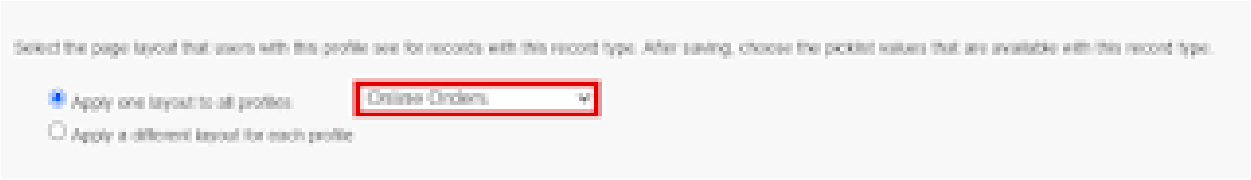
3. Enter Details :
Existing Record Type : Master
Record Type Label : Online Order
Record Type Name : Auto Populates

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4. In Assign Page Layouts
    - Select Apply one layout to all profiles : select **Online orders** from the drop down menu.



5. Click **Save and New**.

Activity 2: Walk-In Orders Record Type

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the Record Types → click New.

3. Enter Details :
Existing Record Type : Master
Record Type Label : Walk-In Order
Record Type Name : Auto Populates
Active : Tick the checkbox
Click Next

4. In Assign Page Layouts
    - Select Apply one layout to all profiles : select **Walk-In orders** from the drop down menu.

5. Click **Save.**

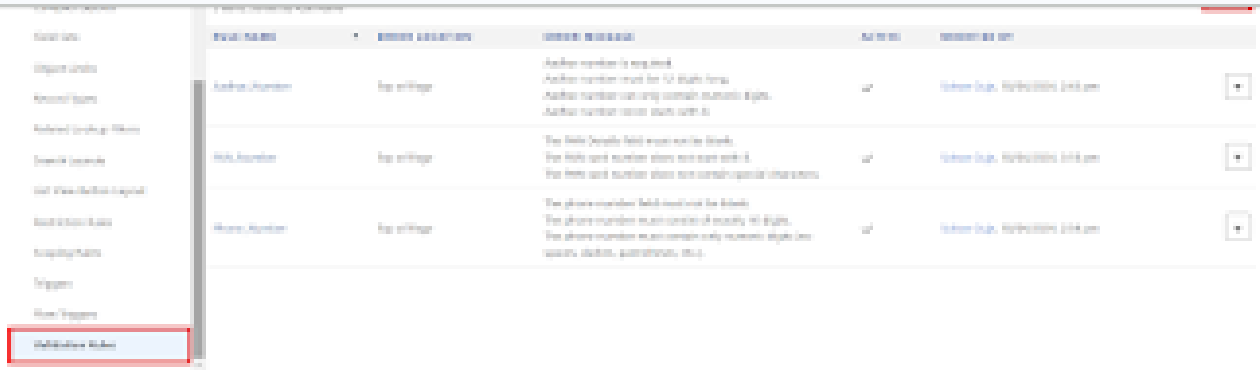
Milestone 9 - Validation Rules

Activity 1: On Employer Account

1. Go to the setup page → click on object manager → From drop down click edit for **Employer Account** object.

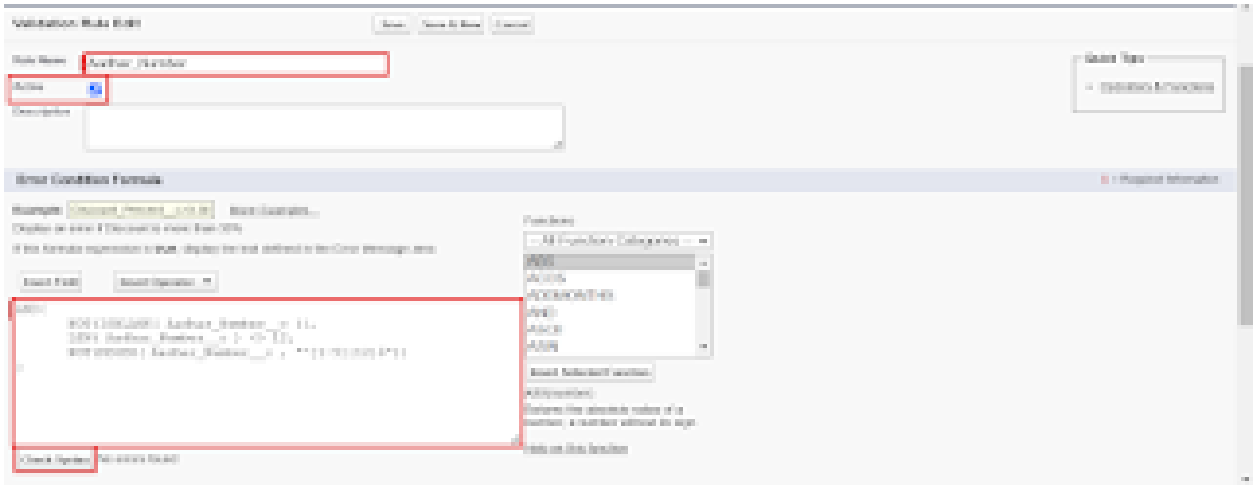
2. Click on the validation rule → click New.

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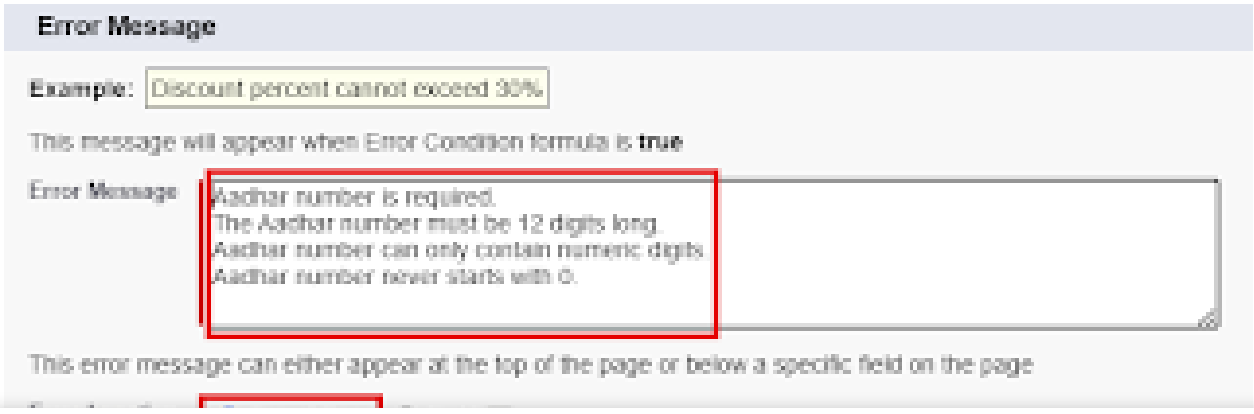
3. **Enter Rule Name :** Aadhar Number
Error Condition Formula :

```
AND(
NOT(ISBLANK( Aadhar_Number__c )),
LEN( Aadhar_Number__c ) <> 12,
NOT(REGEX( Aadhar_Number__c , "^[1-9]{12}$"))
)
```



4. **Error Message** : Aadhar number is required.
The Aadhar number must be 12 digits long.
Aadhar number can only contain numeric digits.
Aadhar number never starts with 0

5. Click **Save & New.**



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```
NOT(REGEX( Phone_Number__c , "^[1-9]{10}$")),
NOT(REGEX( Phone_Number__c , "^\\d+$"))
)
```

Error Message : The PAN Details field must not be blank.
The PAN card number does not start with 0.
The PAN card number does not contain special characters.

Click **Save & New.**

- **Enter Rule Name** : Phone Number

Error Condition Formula :

```
AND(
    NOT(ISBLANK( Phone_Number__c )),
    NOT(REGEX( Phone_Number__c , "^[1-9]{10}$")),
    NOT(REGEX( Phone_Number__c , "^\\d+$"))
    )
```

Error Message :
The phone number field must not be blank.
The phone number must consist of exactly 10 digits.
The phone number must contain only numeric digits
(no spaces, dashes, parentheses, etc.)

Click **Save.**

Activity 2: On Customer Account

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the validation rule → click New.

3. **Enter Rule Name** : Phone Number

Error Condition Formula :

```
AND(
    NOT(ISBLANK( Phone_Number__c )),
    NOT(REGEX( Phone_Number__c , "^[1-9]{10}$")),
    NOT(REGEX( Phone_Number__c , "^\\d+$"))
    )
```

Error Message :
The phone number field must not be blank.
The phone number must consist of exactly 10 digits.
The phone number must contain only numeric digits
(no spaces, dashes, parentheses, etc.)

4. Click **Save.**

Activity 3: On Enterprise Account

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```
NOT(REGEX( Phone_Number__c , "^\\d+$"))
)
```

Error Message :

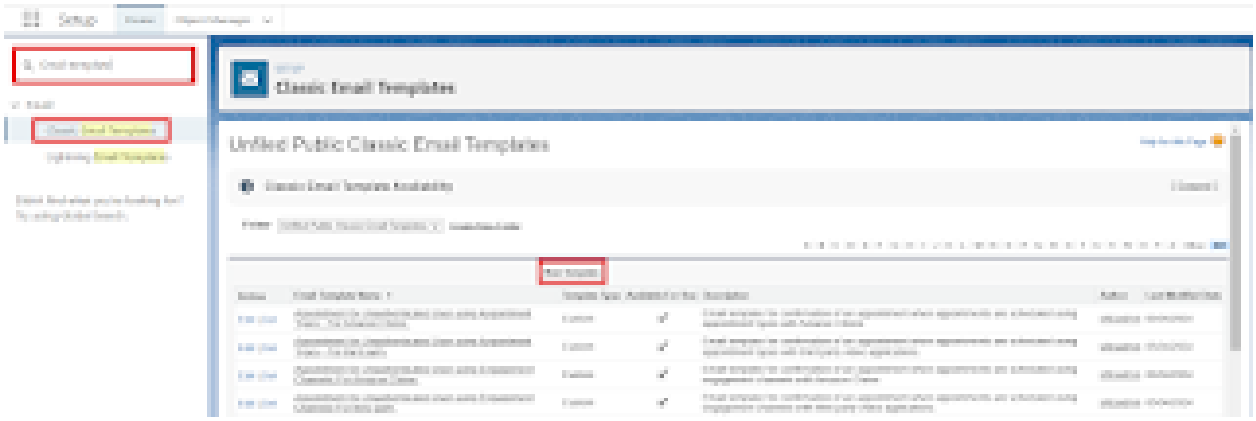
The phone number field must not be blank.
The phone number must consist of exactly 10 digits.
The phone number must contain only numeric digits
(no spaces, dashes, parentheses, etc.)

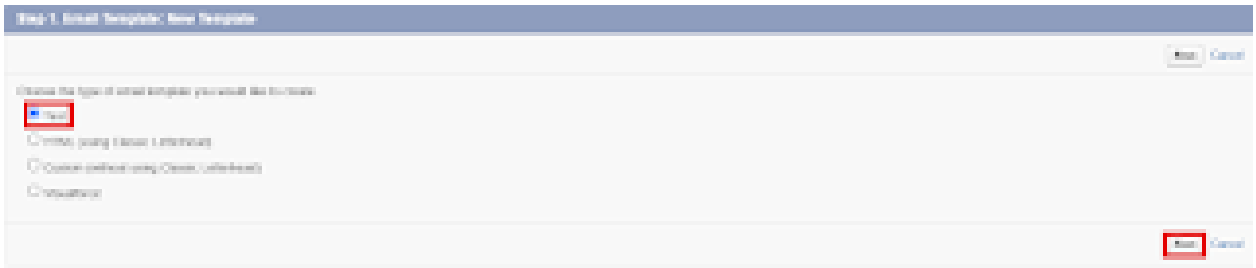
4. Click **Save.**

Milestone 10 - Email Template

Activity 1: Email Template

1. Go to setup, In quick find box enter Email template → click on classic Email Template.
2. Click on→ New Email Template
3. Choose the type of email template you would like to create : **Text** and click **Next**.





- **Available Merge Fields**

Select Field Type : Customer Accounts Fields
Select Field : First Name
Copy Merge Field value : Auto Populates, Copy the value and paste it in the email body where required.



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delighted to offer you an exclusive birthday discount.

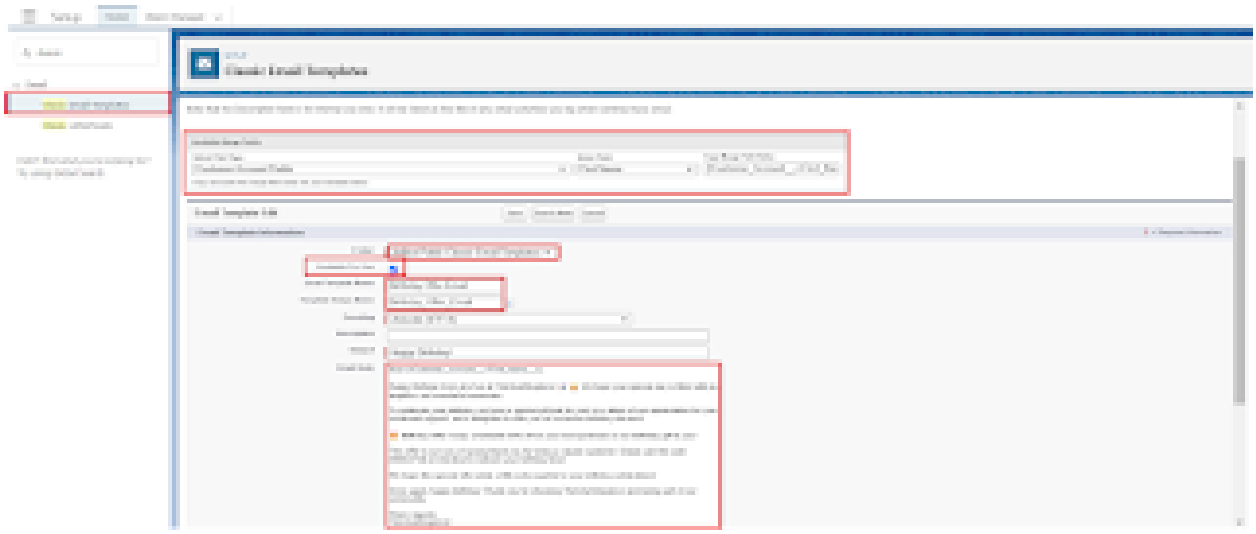
🎁 Birthday Offer: Enjoy a fantastic 50% off on your next purchase as our birthday gift to you!

This offer is our way of saying thank you for being a valued customer. Simply use the code [HBDAY50] at checkout to redeem your birthday treat.

We hope this special offer adds a little extra sparkle to your birthday celebrations!

Once again, happy birthday! Thank you for choosing TheChaiShop&Co! and being part of our community.

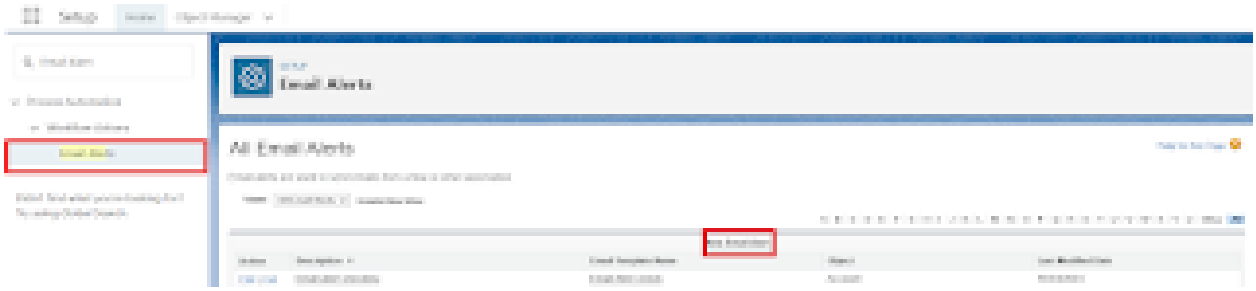
Warm regards
TheChaiShop&Co!



10. Click **Save.**

Activity 2: Email Alert

1. Go to setup, In quick find box enter Email Alert → click on Email Alert .

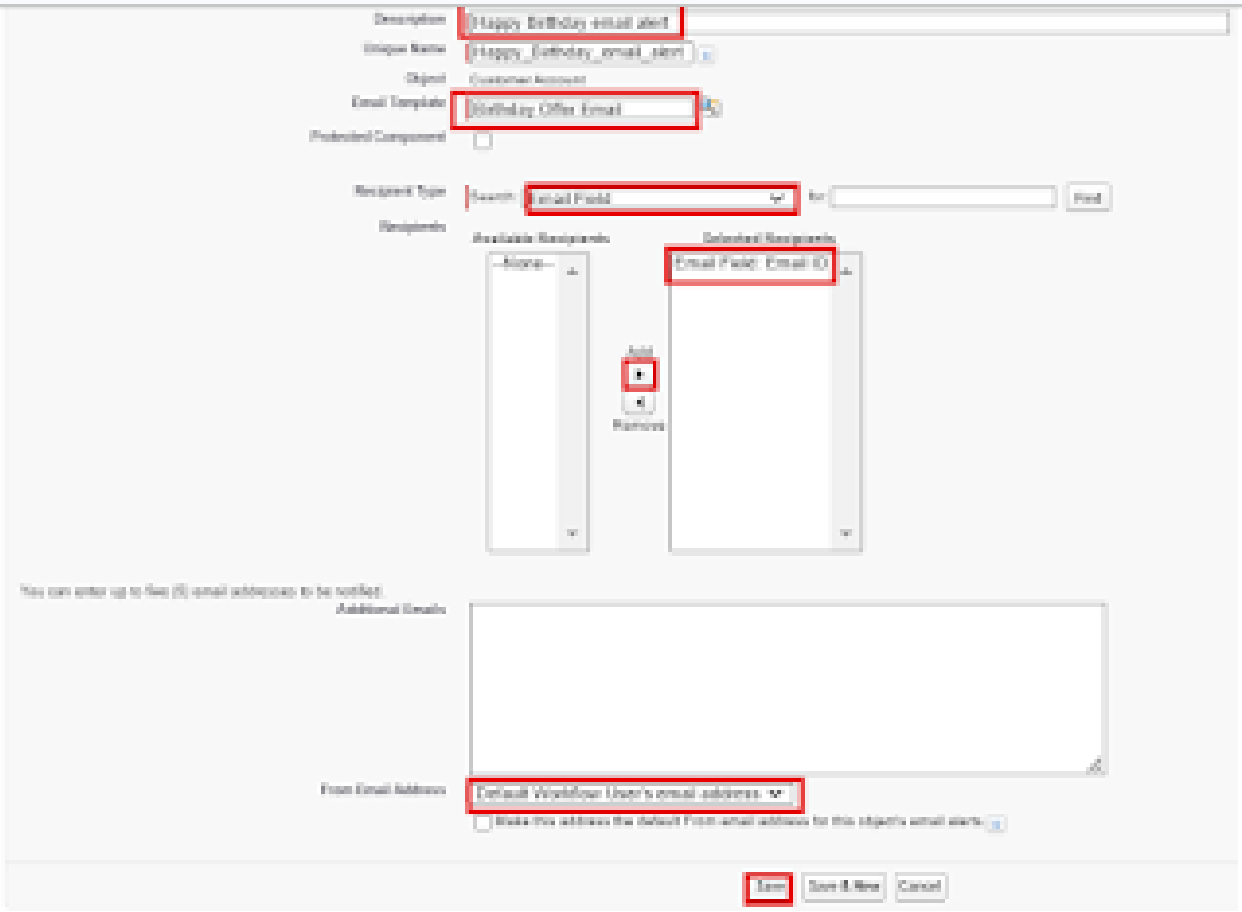


2. Click on New Email Alert.

3. **Description** : Happy Birthday email alert
Unique Name : Auto Populates
Object : Customer Account
Email Template : Search for **Birthday Offer Email**
Recipient Type : Search : Email field : Email ID
From Email Address : Default Workflow user's email address

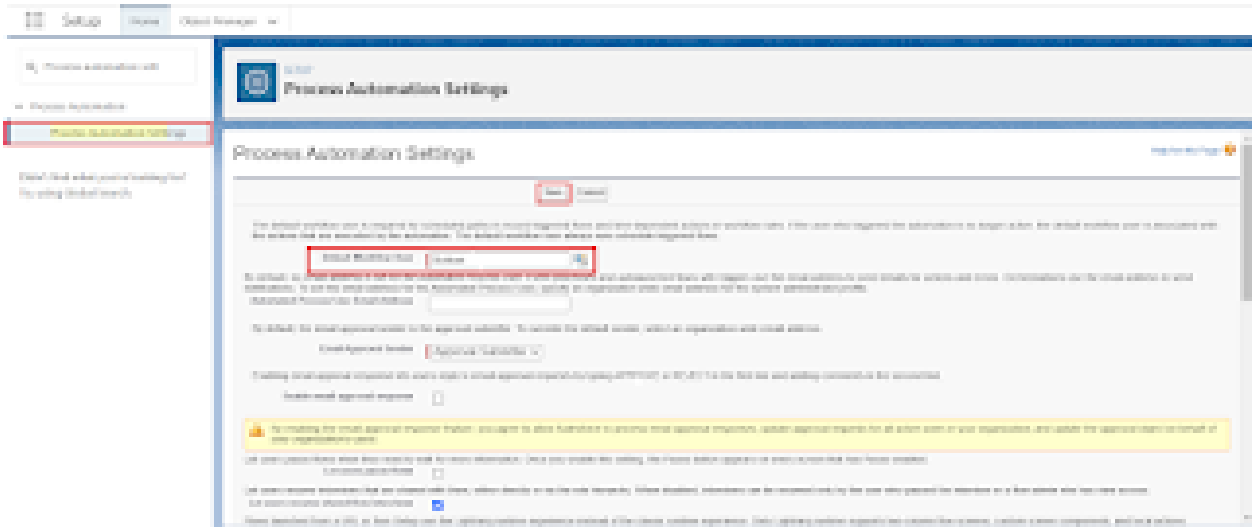
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NOTE :

1. Make sure that the Default workflow user is assigned.

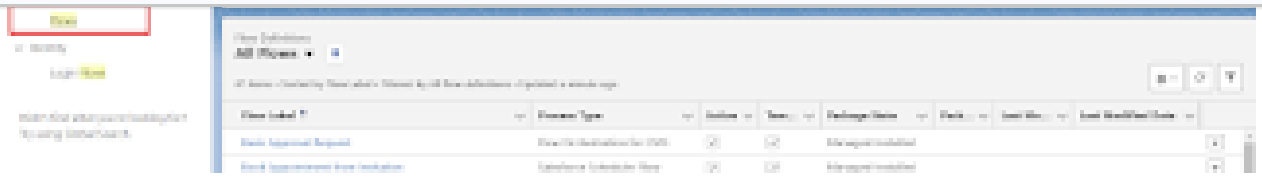


Milestone 11 - Flows

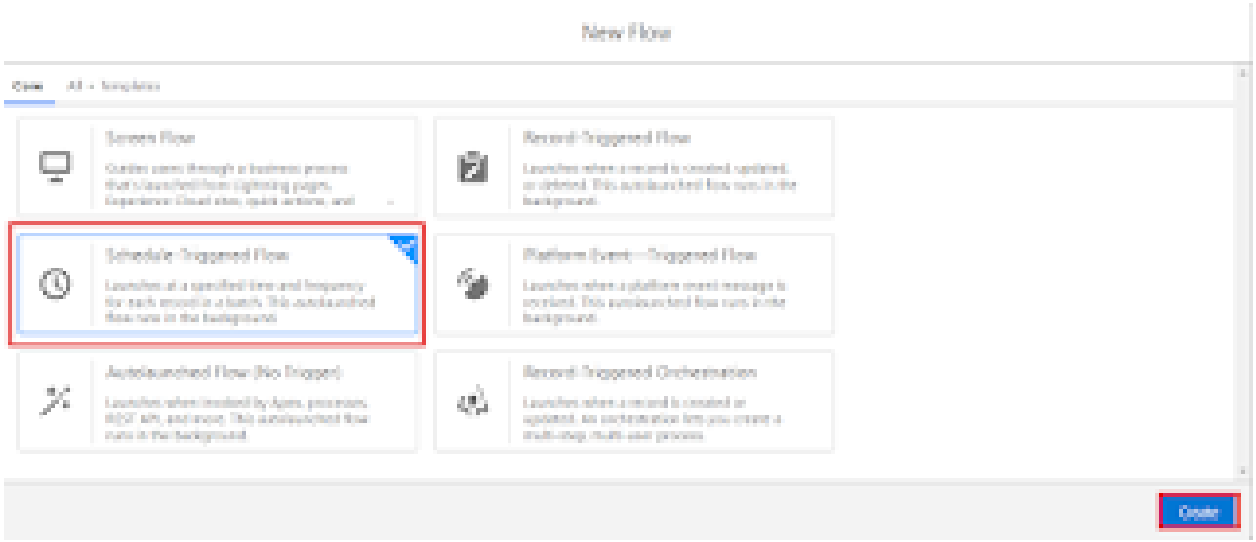
Let's create a scheduled Triggered flow to automate the Birthday offer mails.

Activity 1: Schedule Triggered Flow

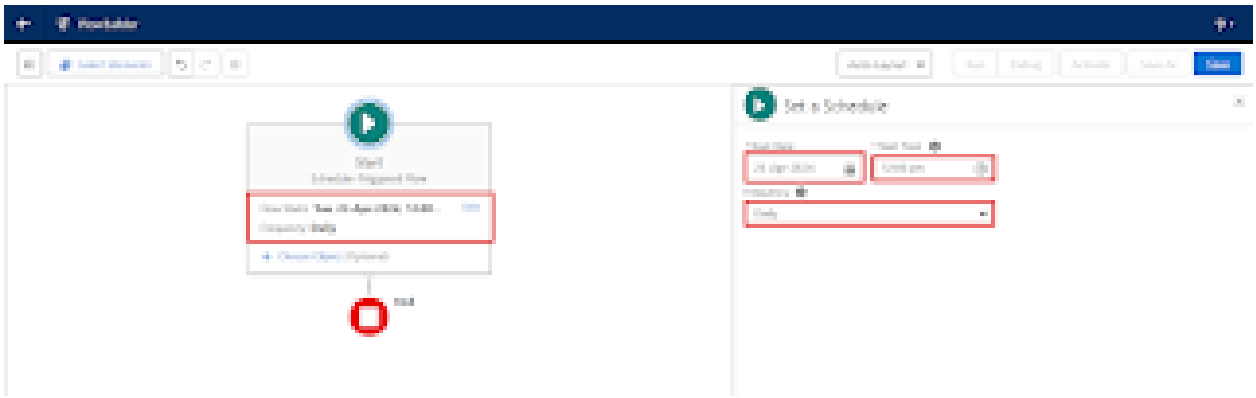
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2. Select Schedule - Triggered Flow and click Create.



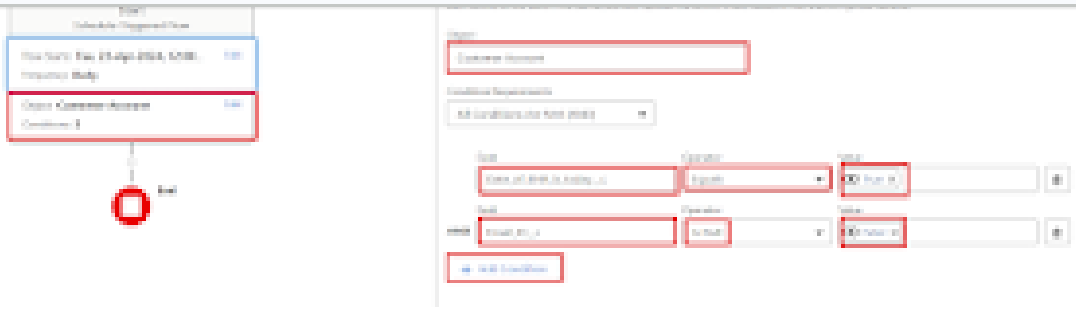
3. **Set a Schedule**
Start Date : Enter the Date, when you want to start the flow.
Start Time : Enter the Time, when you want to start the flow.
Frequency : Daily



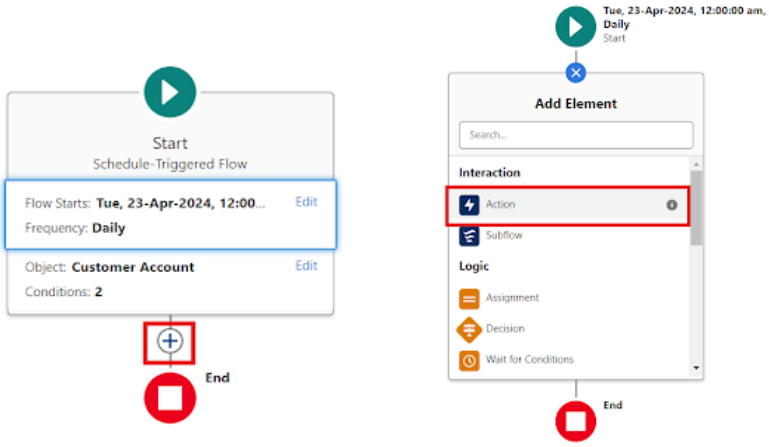
4. **Create Object**
Object : Customer Account
Condition Requirements : All conditions Are Met (AND)
- Field : Date_of_Birthday_is_today__c
Operator : Equals
Value : $GlobalConstant.True
- Field : Email_ID_c
Operator : IS NULL
Value : $GlobalConstant.False

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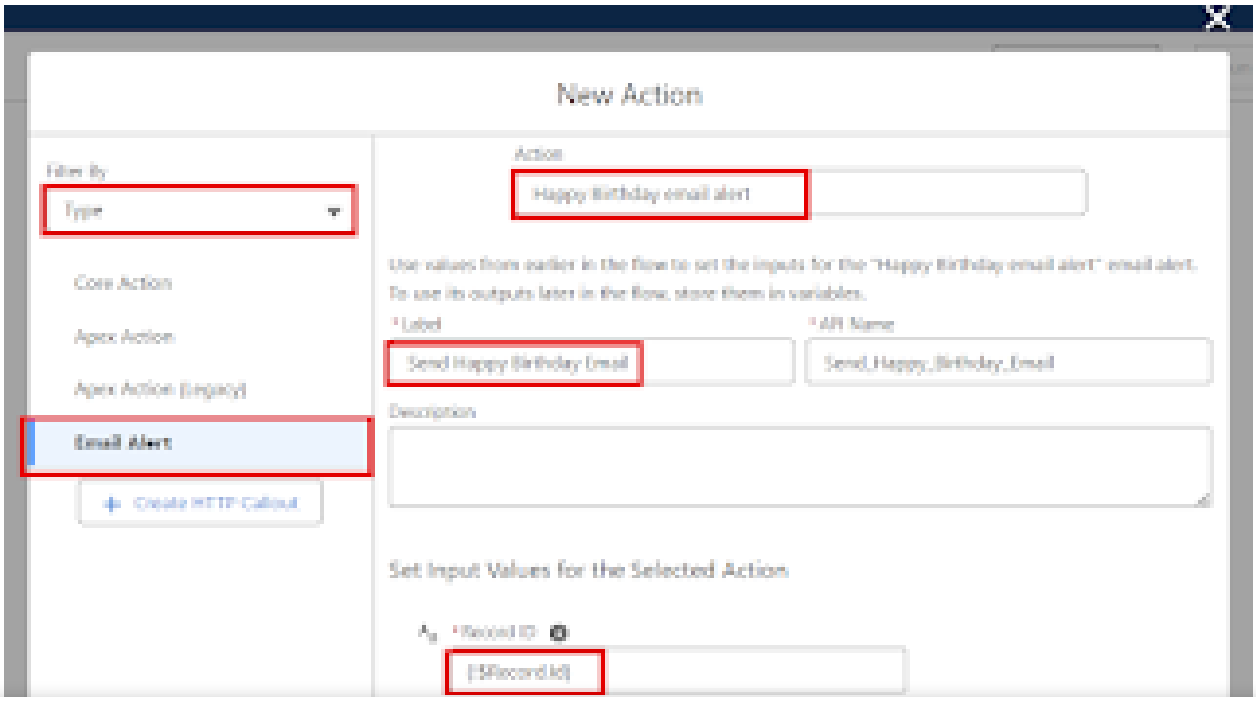
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5. Click (+) Icon on the flow builder and Select **Action Component.**

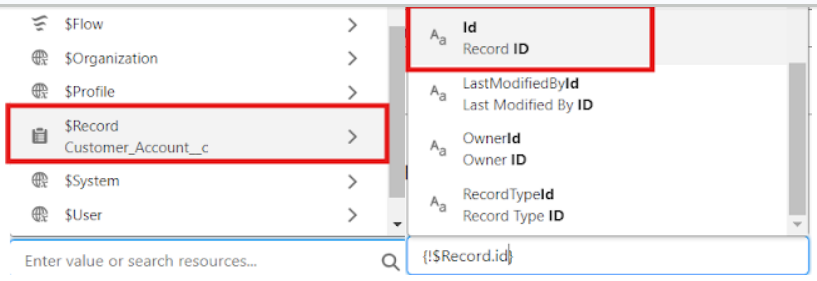


6. In New Action Page
Filter By : Type and select Email Alert
In Action : Search and select Happy Birthday email alert
Label : Send Happy Birthday Email
API Name : Auto Populates
Set Input Values for the Selected Action : Record ID : {!$Record.Id}



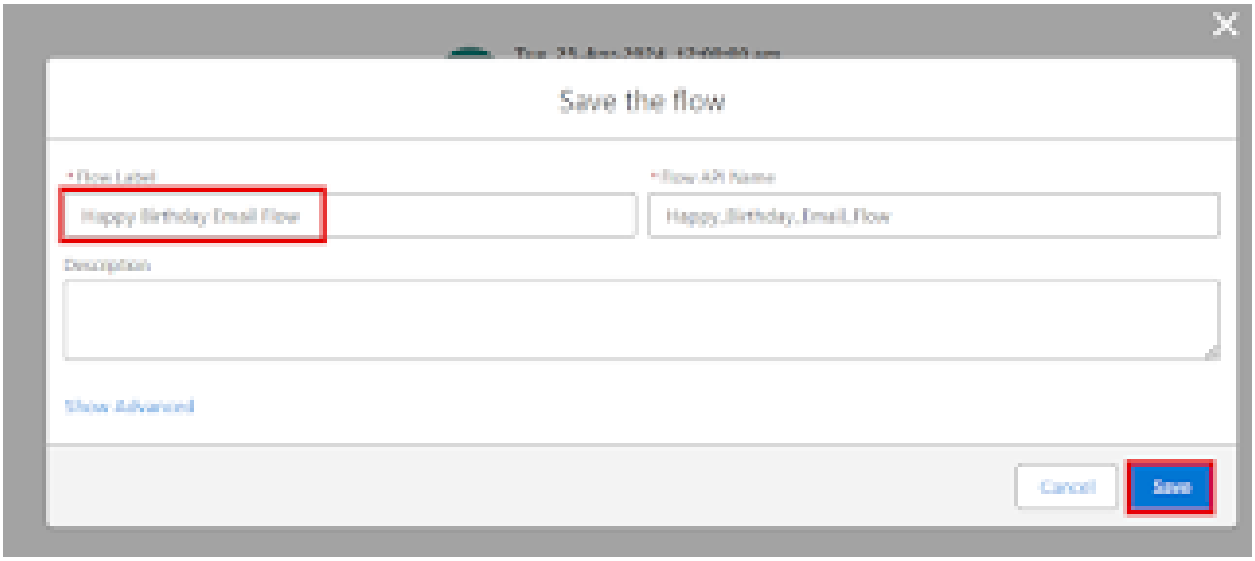
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7. Click **Done**.

8. Click Save on the top right
Flow Label : Happy Birthday Email Flow
Flow API Name : Auto Populates
Click **Save.**



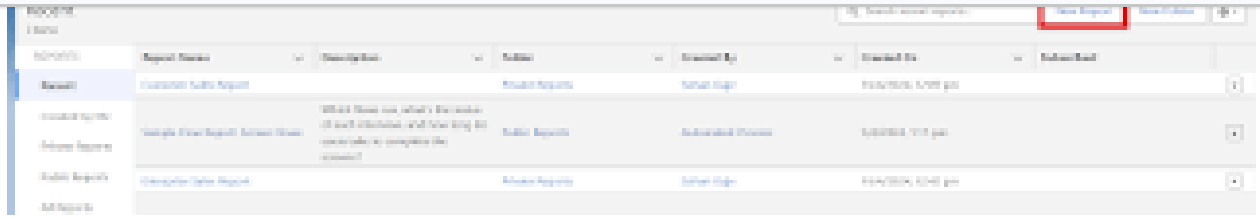
9. Click **Activate**.



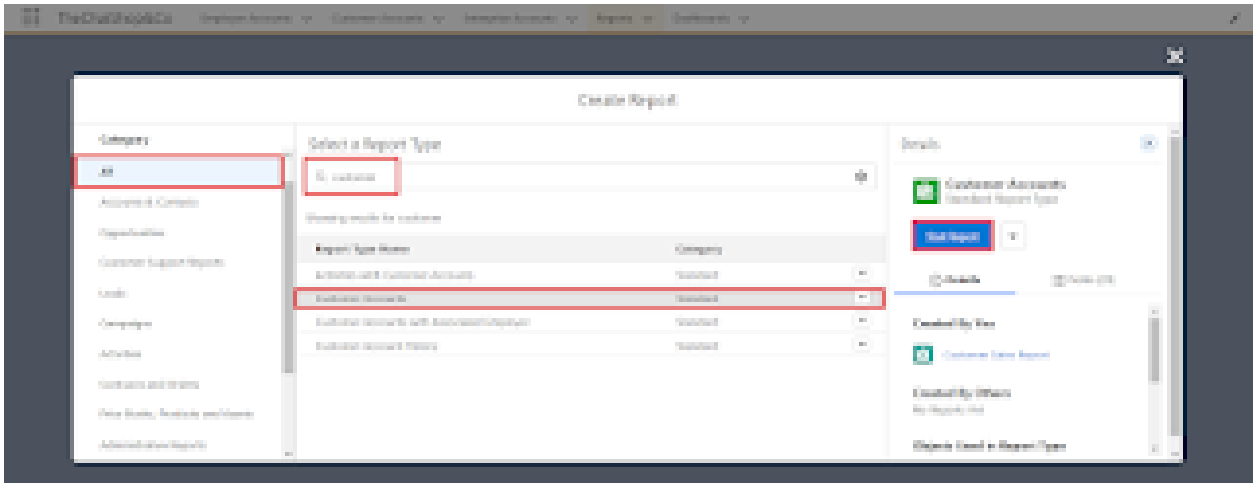


Milestone 12 - Reports & Dashboards

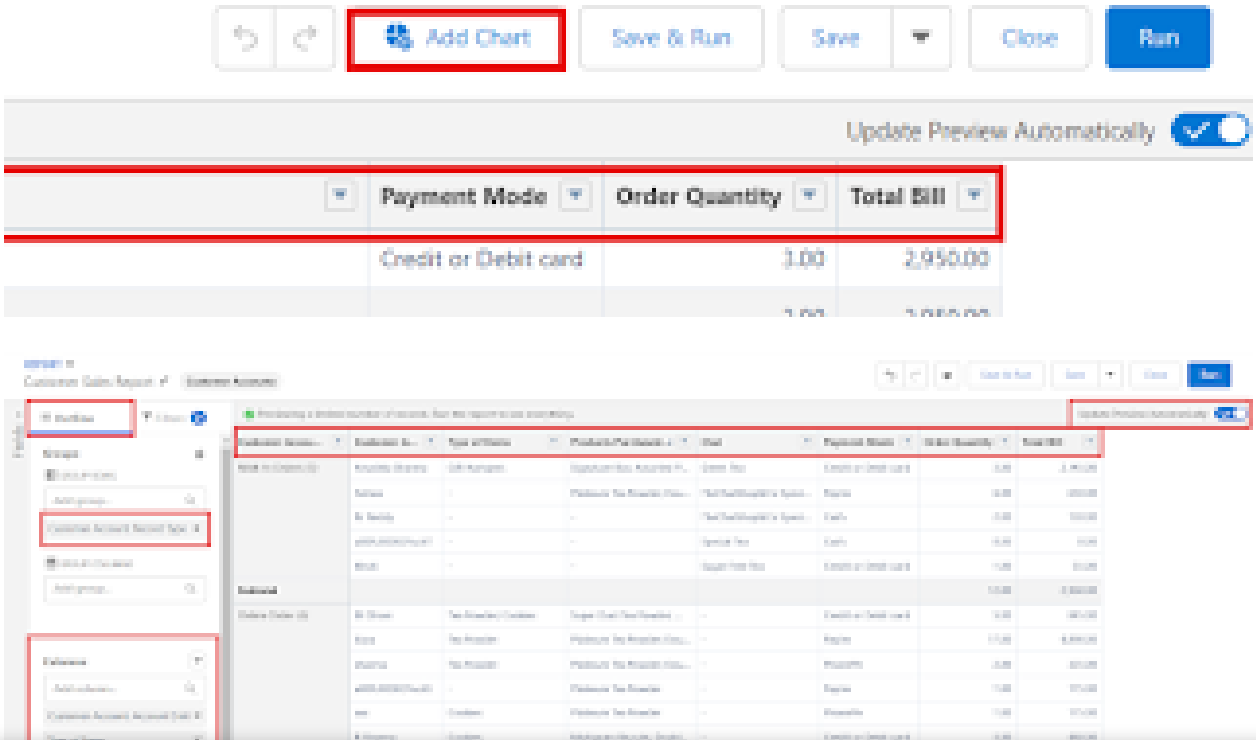
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4. In search type Customer Accounts and Select Customer Accounts.
5. Click Start Report.



6. Click the toggle **Update Preview Automatically** to **ON**.
7. Drag and Drop all the required Columns from the Outline pane.
8. **Group Rows** : Search and Select
   Customer Account: Record Type
9. Click Add Chart on the top.



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Make tweaks, leave comments and share with others to edit at the same time.

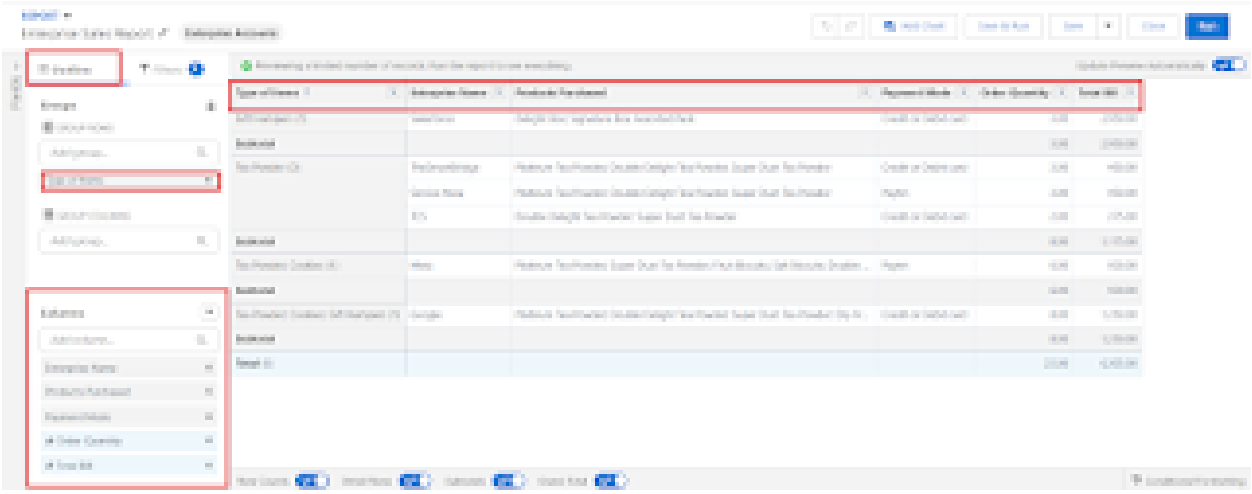
NO, THANKS
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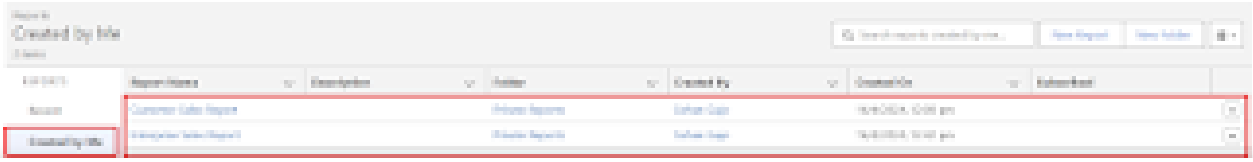
Click **Save**.

Activity 2: Enterprise Account Sales Report

1. Repeat the same steps to create another report.
2. Click Reports
3. Click New Report
4. In search type Enterprise Accounts and Select Enterprise Accounts.
5. Click Start Report.
6. Click the toggle **Update Preview Automatically** to **ON**.
Drag and Drop all the required Columns from the Outline pane.
Group Rows : Search and Select
Type of Items.
7. Click Add Chart on the top.

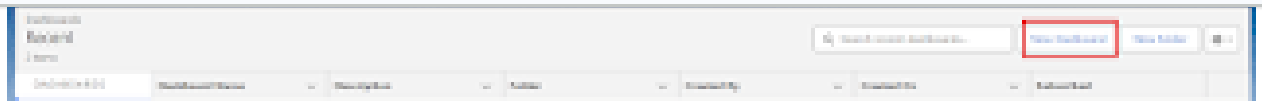


8. Click **Save** on the top right.
    a. Enter
    b. Report Name : Enterprise Account Sales Report
    c. Report Unique Name : Auto Populates
    d. Click **Save**.

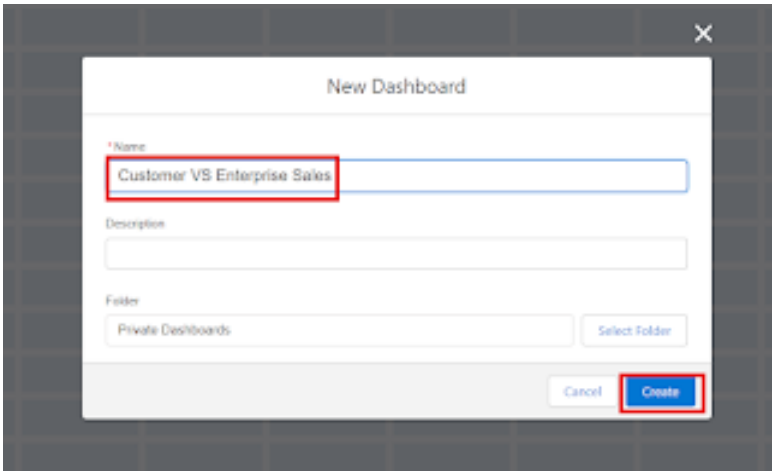


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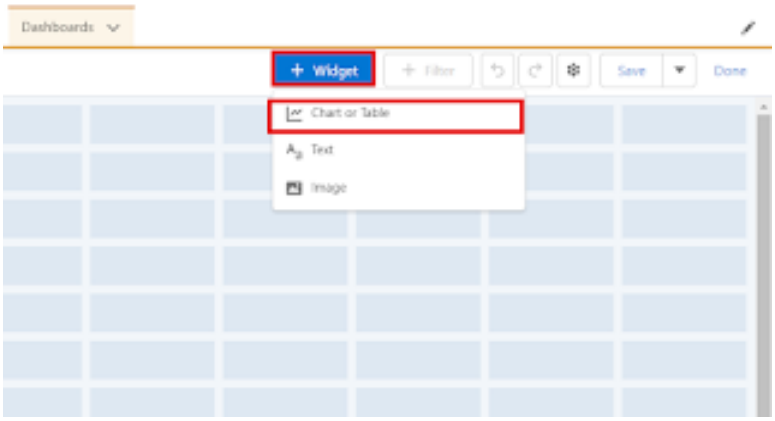
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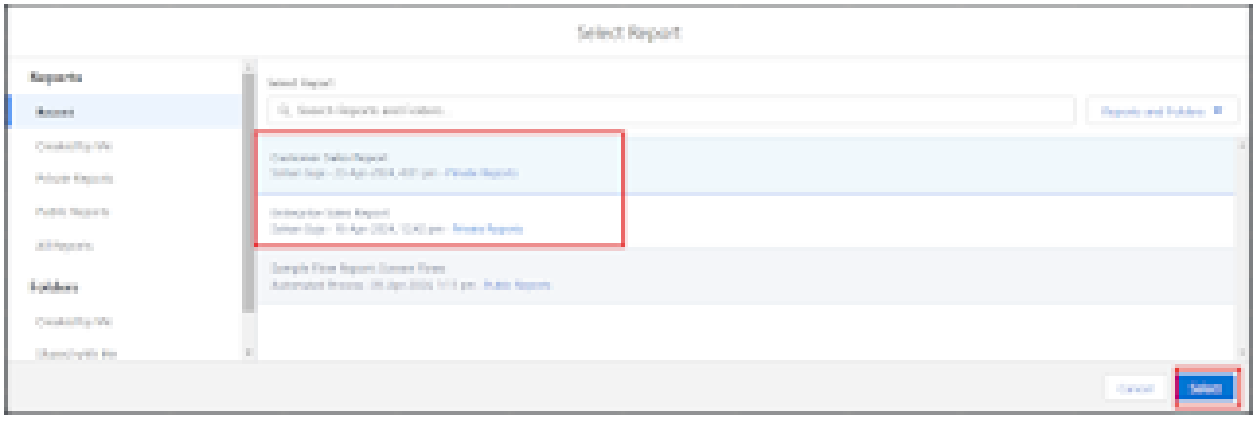
3. Name : Customer Vs Enterprise Sales
4. Click Create.



5. Click Widget
6. Select Chart or Table.



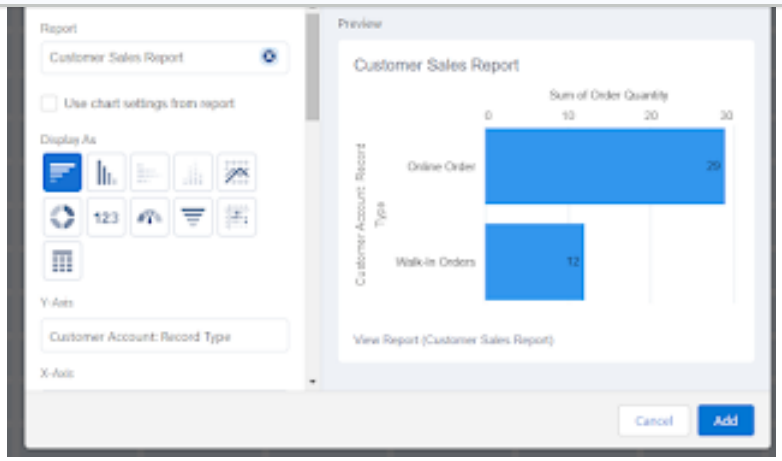
7. Select Customer Account Sales Report



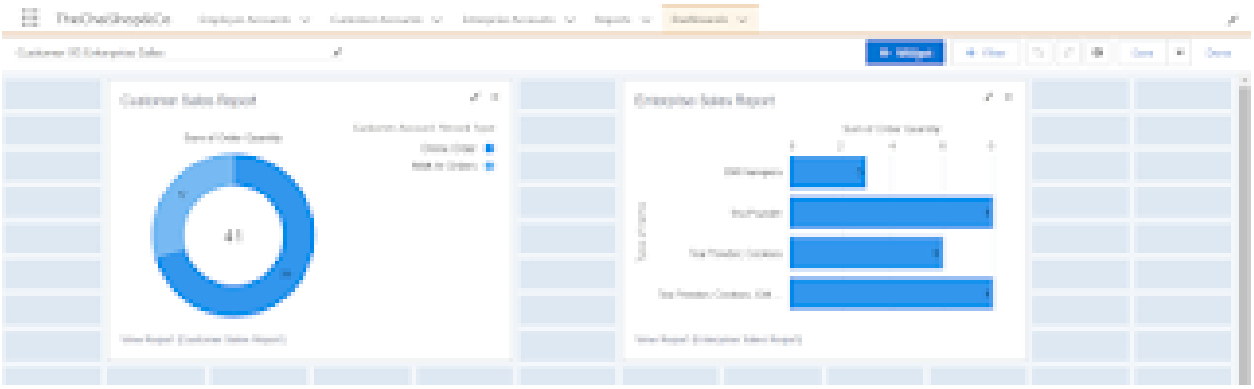
8. Display As : Select the one which you feel comfortable with.
9. Click **Add**.

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10. Repeat the same steps for the Enterprise Account Sales Report.
11. Click **Save**.



Milestone 12 - Conclusion

In conclusion, the development of TheChaiShop&Co CRM application marks a significant stride in enhancing the store's operational efficiency and customer relationship management capabilities. Seamlessly integrating employer details, customer information (both online and offline), and enterprise data, this application streamlines business processes and enables informed decision-making. The automation of sales calculations, including order quantity and total bill amount, reduces errors and empowers employees to focus on delivering exceptional service. Moreover, automated email notifications for customers' birthdays foster personalized communication and enhance customer engagement.

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- Record creation for Employer, Customer, and Enterprise objects
- Validation rule testing for Aadhar, PAN, and phone numbers
- Verification of record types and page layouts
- Formula field validation for order quantity and billing
- Flow execution for birthday email alerts
- Report and dashboard generation

Performance testing ensured fast response time and system stability during concurrent usage.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data management
- Reduced manual work
- Automated calculations
- Improved customer engagement
- Real-time analytics

Disadvantages

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9. Conclusion:

The CRM Application for TheChaiShop&Co successfully improves business operations by integrating customer data, automating billing, and enabling intelligent reporting. The system reduces errors, saves time, and enhances customer relationships through automation and analytics.

10. FUTURE SCOPE

- Mobile CRM integration
- AI-based sales prediction
- WhatsApp/SMS alerts
- Inventory management
- Multi-branch support
- Advanced customer profiling

11. APPENDIX

Source Code : Salesforce Configuration & Flows

Dataset : Customer and Sales Records

Tools Used : Salesforce Developer Org

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